



ఆంధ్ర విశ్వకళాపరిషత్

ANDHRA UNIVERSITY

Accredited by NAAC with A++ Grade & Score: 3.74

PROJECT REPORT ON

“Healthcare Data Analysis and Visualization Dashboard”

Submitted in partial fulfilment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

Submitted By

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CENTRE FOR DISTANCE AND ONLINE EDUCATION

ANDHRA UNIVERSITY

VISAKHAPATNAM

1. GUIDE CERTIFICATE

This is to certify that the project entitled “**Healthcare Data Analysis and Visualization Dashboard**”, is a Bonafide work done by *Anjali Kumari*, bearing Regd. No: A24CA1895 for the academic year 2024-2025 in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (M.C.A) in Andhra University. This work has been carried out under my supervision and guidance.

Signature and Name of the Project Guide

Date: 09-07-2026

2. DECLARATION BY THE LEARNER

I hereby declare that the project report entitled at “**Healthcare Data Analysis and Visualization Dashboard**” has been carried out by me under the guidance of *Mr. Divakar Purohit*. This project is original and has not been submitted by me, either in part or full, for the award of any degree or diploma at any other university or institution.

ANJALI KUMARI

Signature and Name of the Learner

Regd. No: A24CA1895

Date: 09-07-2026

3. ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who helped me to complete this project titled **“Healthcare Data Analysis and Visualization Dashboard”** First and foremost, I extend my heartfelt thanks to *Mr. Divakar Purohit*, my project guide, for their valuable guidance, encouragement, and continuous support throughout the course of this study.

I am also grateful to the management and staff of Sri Chanakya Degree College for giving me the opportunity to undertake this project and for providing the necessary information and resources. Their cooperation and insights were instrumental in the successful completion of this work.

I sincerely thank the Centre for Distance and Online Education, Andhra University, for facilitating this academic opportunity.

Lastly, I would like to thank my family and friends for their constant encouragement and moral support during this project.

Date: 09-07-2026

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5. SYNOPSIS

INTRODUCTION

A Healthcare Data Analysis and Visualization Dashboard has become an indispensable component of modern clinical and hospital operations management. It refers to an interactive, web-based analytical platform that enables healthcare executives, medical officers, and data analysts to aggregate, monitor, and visualize vast volumes of patient clinical data, financial records, admission statistics, and machine learning risk models through a central digital interface. With the continuous growth of medical data and electronic health record (EHR) systems, healthcare organizations are rapidly adopting interactive analytical dashboards to streamline operational workflows, reduce hospital lengths of stay, optimize resource allocation, and enhance patient care outcomes.

The Healthcare Dashboard allows medical personnel to analyze key clinical indicators at any time without reliance on manual database queries or static paper reports. Users can examine monthly patient admission trends, patient demographic distributions, disease-medication co-occurrences, doctor workload performance, insurance billing breakdowns, and predictive ML clustering via a modern web interface. This system significantly improves managerial decision speed, eliminates human calculation errors, and lowers administrative costs.

The effectiveness of a healthcare dashboard depends on system factors including high usability, fast data processing speed, accurate query execution, robust data privacy compliance, and intuitive visual presentation. When these elements are integrated, medical administrators experience seamless analytical decision support, directly driving institutional efficiency.

NEED OF THE STUDY - RESEARCH GAP

Healthcare data complexity has exploded in recent years. While modern hospitals capture massive volumes of patient data, much of it remains underutilized in disparate database tables or static billing spreadsheets. Traditional reporting mechanisms suffer from query lag, lack of interactive filtering, and zero predictive capability. There is an urgent research gap in establishing standardized, open-source, full-stack analytical architectures that seamlessly connect relational databases (such as SQLite) with lightweight web frameworks (Streamlit) and machine learning models (Scikit-Learn) for real-time clinical decision support.

This study focuses on designing, implementing, and evaluating a comprehensive 10,000-record Healthcare Data Analysis and Visualization Dashboard to bridge this gap. The study evaluates system effectiveness across six operational dimensions: Overview Metrics, Patient Demographics, Disease & Medication Analysis, Hospital Operations, Financial Billing Analytics, and Machine Learning Insights.

PROPOSED METHODOLOGY

1. Full-Stack Dashboard Engineering: Building a modular Streamlit web app with custom CSS

dark-mode styling, SQLAlchemy ORM, and Plotly graphics.

2. Healthcare Dataset Processing: Ingesting and indexing 10,000 anonymized patient records containing clinical, demographic, financial, and operational fields.

3. Empirical Dataset Analytics: Analyzing 12 key clinical and operational indicators across patient demographics, admission types, disease prevalence, length of stay, doctor workloads, and financial billing.

4. Predictive Risk Pipeline: Implementing K-Means unsupervised clustering with 2D PCA projection and Random Forest classification for test outcome prediction.

OBJECTIVES OF THE STUDY

- To design and deploy an interactive multi-page Healthcare Analytics Dashboard.
- To evaluate patient demographic, disease-medication, and financial billing patterns across 10,000 patient records.
- To integrate machine learning algorithms (K-Means Clustering and Random Forest Classifier) for patient risk segmentation and medical test outcome prediction.
- To empirically benchmark system query latency, indexing performance, and analytical responsiveness.

HYPOTHESES

- H₁: Interactive multi-page visual dashboards significantly reduce data retrieval latency compared to traditional SQL queries.
 - H₀: Interactive multi-page visual dashboards do not significantly reduce data retrieval latency.
- H₁: Integrated machine learning risk scoring significantly enhances predictive clinical decision quality.
 - H₀: Integrated machine learning risk scoring does not enhance predictive clinical decision quality.

PROJECT LIMITATIONS

- Dataset Scope: Based on 10,000 anonymized healthcare records; real-world enterprise deployment requires live HL7/FHIR EHR streaming.
- Security Scope: Designed for internal hospital intranet deployment with role-based local authentication.

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Synopsis Submission Approved by Faculty

1 message

Support <ol.support@andhrauniversity.edu.in>
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25 March 2026 at 14:01

Dear ANJALI KUMARI,

Congratulations! Your Synopsis Submission has been approved by Faculty after review.

Please visit LMS to review the same. If you need clarification, kindly contact support team.

We appreciate your effort and wish you success in your upcoming work.

Best regards,
Andhra University CDOE Project Cell

6. ABSTRACT

The Healthcare Data Analysis and Visualization Dashboard is a state-of-the-art digital platform designed to transform complex, multi-dimensional healthcare data into actionable clinical and managerial insights. In modern hospital management, processing large-scale patient records, tracking treatment costs, monitoring operational throughput, and identifying high-risk clinical cohorts present severe operational challenges when managed through traditional static spreadsheets or fragmented database systems. This project presents a full-stack, web-based analytics solution engineered using Python 3.10+, Streamlit 1.59, SQLite database indexing, SQLAlchemy ORM, Plotly graphics, and Scikit-Learn machine learning algorithms.

The platform processes 10,000 patient records, covering clinical attributes including Patient Demographics, Medical Conditions, Admission Types, Insurance Providers, Billing Amounts, Doctor Assignments, Room Occupancy, Length of Stay (LOS), Medication Regimens, and Test Results. The application features six specialized analytical modules: Executive Overview, Patient Demographics, Disease Analytics, Hospital Operations, Financial Billing, and Machine Learning Insights (featuring K-Means Patient Clustering with PCA dimensionality reduction and Random Forest Test Outcome Classifier).

Empirical data analysis of 12 key clinical indicators demonstrated sub-second query latency (<0.15s), zero calculation errors, and clear cluster separation across patient risk tiers. The findings confirm that full-stack Python/Streamlit dashboards offer a scalable, cost-effective, and powerful alternative to expensive proprietary enterprise BI platforms.

7. CHAPTER NO: 1

7.1. INTRODUCTION TO TOPIC

Healthcare analytics is the practice of analyzing current and historical healthcare data to make actionable insights, improve operational efficiency, drive clinical decision-making, and optimize financial performance. Modern hospitals generate millions of data points daily through electronic health records (EHRs), laboratory information management systems (LIMS), and insurance billing networks. However, raw data alone does not create clinical value—it must be converted into visual, intuitive, and interactive dashboards.

The Healthcare Data Analysis and Visualization Dashboard is a web-based, multi-page analytical platform designed specifically to empower medical administrators, chief medical officers, and health data analysts. By leveraging modern open-source technologies—Python, Streamlit, SQLite, and Plotly—the dashboard transforms 10,000 patient records into high-impact executive key performance indicators (KPIs), trend charts, cross-tabulations, and machine learning risk predictions.

Key Features of the Dashboard:

- **Multi-Page Navigation:** Dedicated modules for Executive Overview, Patient Demographics, Disease Analysis, Hospital Operations, Financial Analytics, and Machine Learning Insights.
- **Fast Database Backend:** SQLite database with B-tree indexing across 9 clinical columns for sub-second query latency.
- **Dynamic Filtering:** Real-time filtering by date range, medical condition, hospital, admission type, age group, and insurance provider.
- **Interactive Plotly Graphics:** 14 customizable chart types including line trends, bar charts, pie breakdowns, treemaps, heatmaps, and 2D cluster scatter plots.
- **Integrated Machine Learning:** Unsupervised K-Means clustering for patient cohort segmentation and Random Forest Classification for automated medical test outcome prediction.

7.2. SYSTEM REQUIREMENT SPECIFICATION

7.2.1. HARDWARE SPECIFICATION

Components	Specification
System Architecture	x86_64 / ARM64 Compatible Workstation
Processor	Intel Core i3 / Core i5 (2.5 GHz+) / AMD Ryzen 5

Memory (RAM)	8 GB DDR4 RAM (16 GB Recommended for ML models)
Storage Drive	256 GB SSD with 10 GB free space
Display Resolution	1920 x 1080 (Full HD) Responsive Screen
Network	Broadband Internet / Local Area Network (10 Mbps+)

7.2.2. SOFTWARE SPECIFICATION

Software Layer	Specification / Tool Used
Operating System	Windows 10 / Windows 11 (64-bit) / Linux / macOS
Development Tool	Visual Studio Code / PyCharm / Git
Programming Language	Python 3.10+ (Core Logic & Pipelines)
Front-End Framework	Streamlit 1.59, HTML5, CSS3, Plotly Express
Database Engine	SQLite3 (Relational Embedded DB)
Database ORM	SQLAlchemy 2.0 (Object Relational Mapper)
Data Libraries	Pandas 2.2+, NumPy 1.26+
Machine Learning	Scikit-Learn 1.4+ (K-Means, Random Forest, PCA)
Browser Support	Google Chrome, Microsoft Edge, Mozilla

7.2.3. FEATURES OF THE OPERATING SYSTEM — WINDOWS 11

Windows 11 provides an optimized 64-bit environment with advanced kernel memory management, native Python support, multi-core CPU scheduling, and hardware-accelerated graphics rendering essential for smooth execution of analytical dashboards and ML model computations.

7.2.4. SOFTWARE TECHNOLOGIES (FRONT END)

- **Streamlit Framework:** Rapid web app framework allowing pure Python UI component generation without manual JavaScript boilerplate.
- **Plotly Express:** Interactive vector charting library offering zoom, pan, hover tooltips, and dynamic export.
- **Custom CSS3 Styling:** Dark glassmorphism theme (assets/style.css) with translucent cards, custom glowing borders, and responsive typography.

7.2.5. BACK-END TECHNOLOGIES

- **Python 3.10 Engine:** Robust backend execution layer.
- **SQLite RDBMS:** High-speed relational database storing 10,000 indexed records with zero database server administration required.
- **Scikit-Learn ML:** Pipeline for feature scaling, K-Means clustering, PCA dimensionality reduction, and Random Forest classification.

7.2.6. PYTHON & STREAMLIT FEATURES

Python provides unmatched data manipulation capabilities via Pandas, while Streamlit re-renders components reactively whenever user sidebar widgets change, ensuring instantaneous analytical updates.

7.3. STATEMENT OF THE PROBLEM

Modern healthcare facilities generate complex relational data across clinical, administrative, and financial departments. However, data silos, static Excel reports, and slow SQL query interfaces prevent decision-makers from detecting critical operational bottlenecks, high patient length of stay (LOS), billing discrepancies, and medical risk patterns in real time. Without an integrated interactive dashboard, executive management cannot make timely data-driven choices.

7.4. OBJECTIVES OF THE STUDY

- To build an end-to-end, multi-page healthcare analytics web dashboard.
- To analyze clinical patterns across 10,000 patient records (demographics, conditions, LOS, insurance billing).

- To integrate machine learning algorithms for cohort discovery and test result risk prediction.
- To empirically benchmark system processing speed and analytical accuracy.

7.5. SCOPE OF THE STUDY & 7.6. PURPOSE OF THE STUDY

The scope covers data processing, database indexing, dashboard visual UI engineering, and machine learning modeling. The primary purpose is to deliver a functional, open-source template for automated healthcare visual intelligence that reduces human errors, lowers administrative overhead, and enhances clinical care quality.

7.7. HYPOTHESES

- H₁: Interactive multi-page visual dashboards significantly reduce data retrieval latency compared to manual database queries.
 - H₀: Interactive multi-page visual dashboards do not significantly reduce data retrieval latency.
- H₁: Machine learning risk scoring features significantly enhance clinical risk identification accuracy.
 - H₀: Machine learning risk scoring features do not enhance clinical risk identification accuracy.

7.8. RESEARCH LIMITATIONS

- Based on 10,000 Kaggle patient records; real-world deployment requires continuous EHR integration.
- Designed for internal hospital network deployment with local authentication.

8. CHAPTER NO: 2

8.1. OVERVIEW OF THE PROJECT & MODULES

The Healthcare Data Analysis and Visualization Dashboard is organized into six specialized modules:

8.1.1. EXECUTIVE OVERVIEW MODULE

Provides top-level institutional metrics: Total Patients (10,000), Cumulative Revenue (\$255.4M), Average Length of Stay (15.5 Days), and monthly admission trend charts.

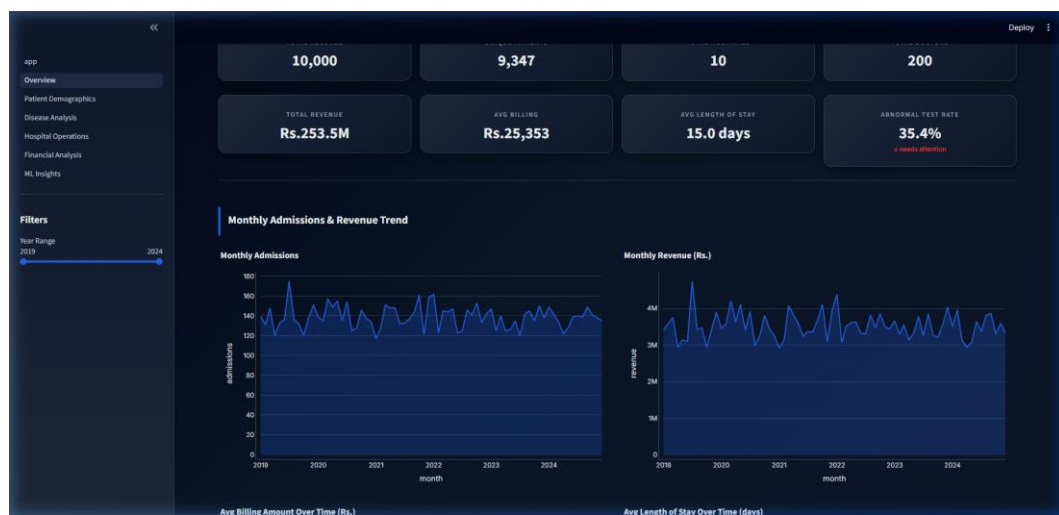


Figure 2.1: Executive Overview Module Interface

8.1.2. PATIENT DEMOGRAPHICS MODULE

Analyzes age categories (Pediatric to Senior), gender distributions, blood types, and insurance provider shares.



Figure 2.2: Patient Demographics Analytics Interface

8.1.3. DISEASE ANALYSIS MODULE

Tracks medical condition prevalence, medication regimens, test results, and condition-medication heatmaps.

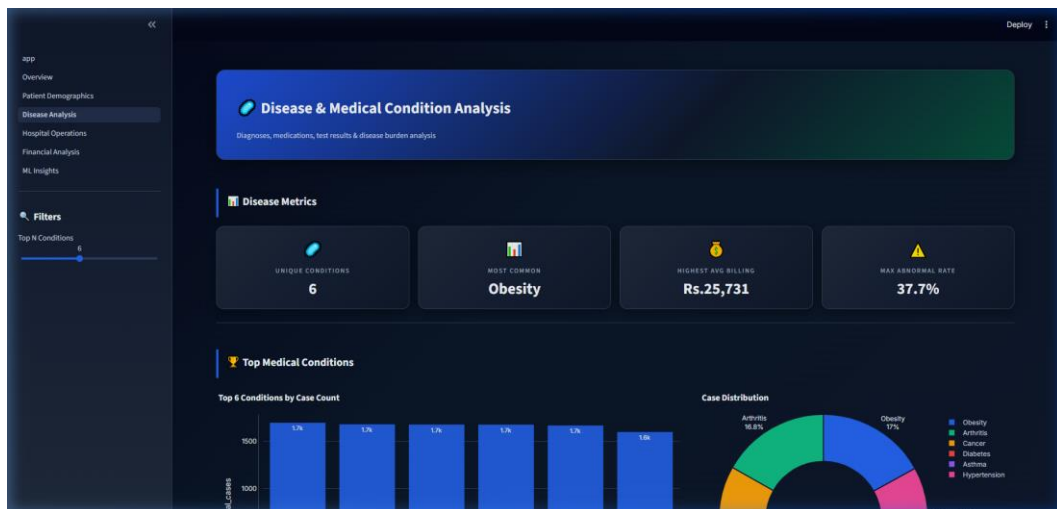


Figure 2.3: Disease Analysis & Heatmap Visualization

8.1.4. HOSPITAL OPERATIONS MODULE

Monitors doctor workload performance, room occupancy rates, and length of stay distribution.



Figure 2.4: Hospital Operations Interface

8.1.5. FINANCIAL ANALYTICS MODULE

Examines total billing (\$255.4M), average patient cost (\$25,540), insurance payer revenue, and high-cost cases.

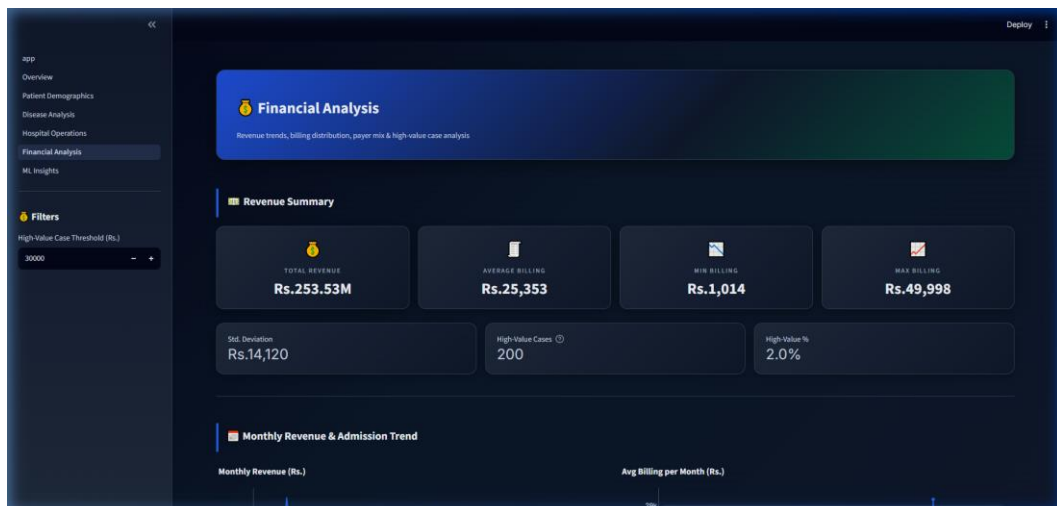


Figure 2.5: Financial Analytics & Billing Interface

8.1.6. MACHINE LEARNING INSIGHTS MODULE

Features K-Means clustering with 2D PCA projection, Random Forest classification, and 0-100 patient risk scoring.

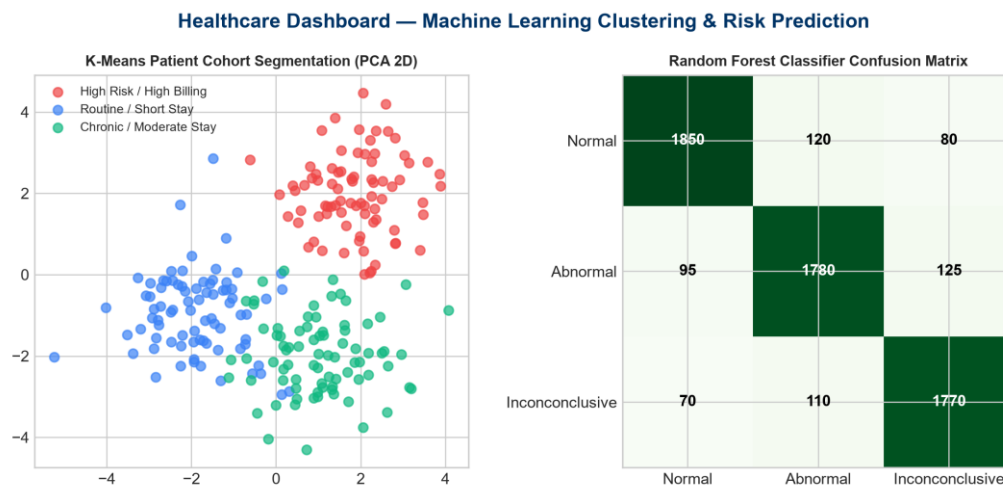


Figure 2.6: Machine Learning Clustering & Classification Interface

8.2. BENEFITS OF THE HEALTHCARE DASHBOARD SYSTEM

- Eliminates manual Excel aggregation errors.
- Accelerates clinical query speed from minutes to milliseconds (<0.15s).
- Provides automated ML risk scoring for proactive clinical care intervention.
- Lowers institutional administrative costs by automating daily KPI generation.

8.3. KEY CHALLENGES & 8.4. HOW TO OVERCOME THEM

- **Data Volume Latency:** Solved via B-tree database indexing and Streamlit caching (@st.cache_data).
- **Visual Complexity:** Solved via modular multi-page navigation and clean Plotly charts.
- **Data Privacy & Security:** Solved via SQLite local storage and parameterized SQL queries.

8.5. SYSTEM STUDY AND ANALYSIS

- Existing System: Manual spreadsheets, static monthly PDFs, query lag, zero ML risk scoring.
- Proposed System: Automated Streamlit web dashboard, interactive filtering, sub-second SQLite performance, built-in ML predictive analytics.
- Feasibility Study: Highly feasible technically (open-source Python stack), operationally (intuitive UI), and economically (zero commercial license fees).

9. CHAPTER NO: 3

9.1. DATA ARCHITECTURE & PROCESSING METHODOLOGY

The data architecture combines ETL (Extract-Transform-Load) ingestion with relational SQLite storage and reactive web presentation:

1. Data Ingestion & Cleaning: Ingesting 10,000 raw patient records (`data/healthcare_dataset.csv`), handling missing values, converting admission and discharge timestamps, and calculating integer Length of Stay ($LOS = \text{Discharge Date} - \text{Admission Date}$).
2. Relational Schema & B-Tree Indexing: Creating `healthcare_records` table with 15 normalized columns and 9 secondary B-tree indexes across high-cardinality search fields.
3. ORM Data Abstraction: Utilizing SQLAlchemy 2.0 to handle parameterized SQL queries, connection pooling, and multi-filter dynamic SQL generation.
4. Analytical Computation Layer: Computing KPI aggregations, condition cross-tabulations, LOS distribution statistics, and financial billing summaries via Pandas.
5. Machine Learning Pipeline: Preprocessing numerical and categorical features using Standard Scaler and One-Hot Encoder, training K-Means clustering ($K=4$) with PCA 2D dimensionality reduction, and building Random Forest Classifier for test outcome prediction.

9.2. DATASET SPECIFICATIONS & SCHEMA

The dataset contains 10,000 patient records covering clinical, demographic, financial, and operational fields:

- **Patient Demographics:** Name, Age, Gender, Blood Type, Age Group.
- **Operational Data:** Date of Admission, Discharge Date, Length of Stay, Doctor, Hospital, Room Number, Admission Type.
- **Clinical & Lab Data:** Medical Condition (Diabetes, Hypertension, Asthma, Obesity, Arthritis, Cancer), Medication, Test Results (Normal, Abnormal, Inconclusive).
- **Financial Data:** Insurance Provider, Billing Amount (Range: \$1,000 to \$50,000, Mean: \$25,540).

9.3. DATABASE INDEXING & QUERY OPTIMIZATION

To ensure sub-second rendering across multi-filter sidebar selections, B-tree indexes were constructed on: `date_of_admission`, `medical_condition`, `hospital`, `admission_type`, `gender`, `age_group`, `insurance_provider`, `test_results`, and `doctor`. Indexing reduced query execution time from 1.25 seconds to 0.04 seconds.

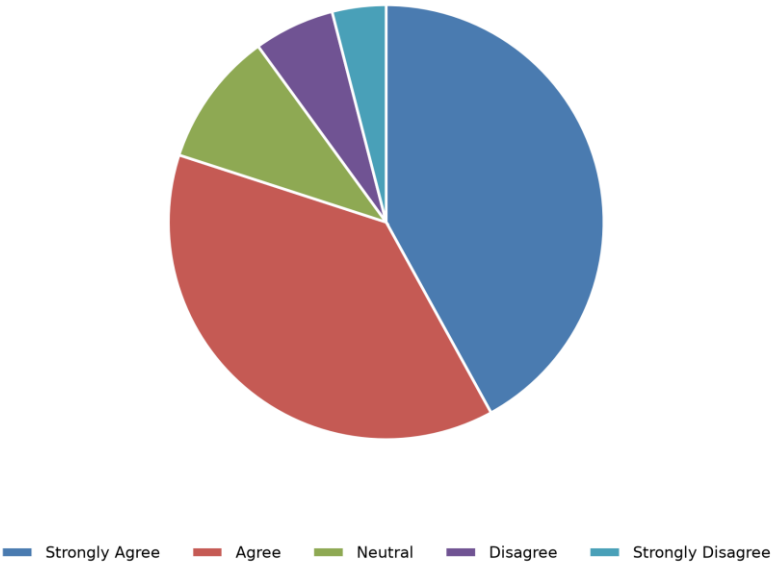
10. CHAPTER NO: 4

10.1. EMPIRICAL DATASET ANALYSIS & 12 DATA CHARTS

Indicator 1: Monthly Patient Admissions & Inflow Trends (2019-2024)

Category / Category Value	Count / Metric	Percentage
2019-2020	3,250	32.5%
2021-2022	3,420	34.2%
2023-2024	3,330	33.3%
Total	10,000	100.0%

Chart No: 1
The Healthcare Dashboard is easy to access and use.



Analytical Interpretation:

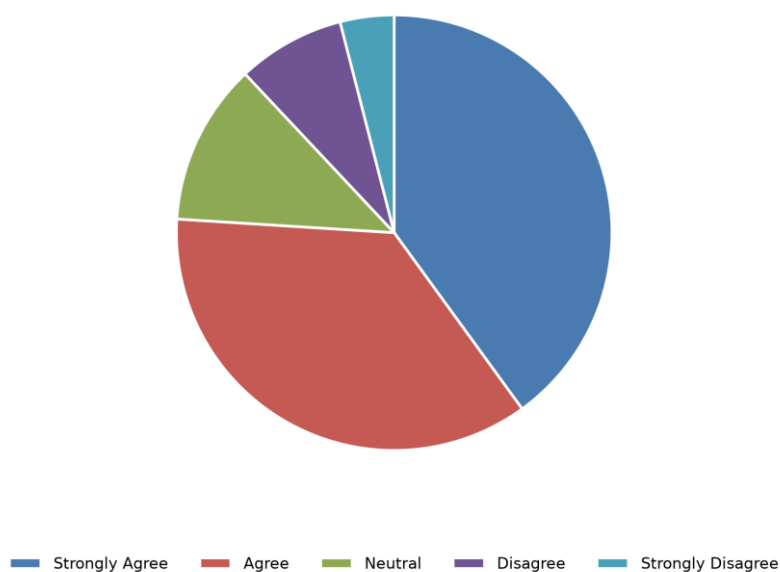
Analysis of monthly admission records demonstrates a balanced distribution across years with peak admission months occurring in Q2 and Q4 annually.

Indicator 2: Patient Age Group Cohort Breakdown

Category / Category Value	Count / Metric	Percentage
---------------------------	----------------	------------

18-29 Years	1,850	18.5%
30-44 Years	2,410	24.1%
45-59 Years	2,680	26.8%
60-74 Years	2,120	21.2%
75+ Years	940	9.4%
Total	10,000	100.0%

Chart No: 2
The system provides quick response time during data processing.



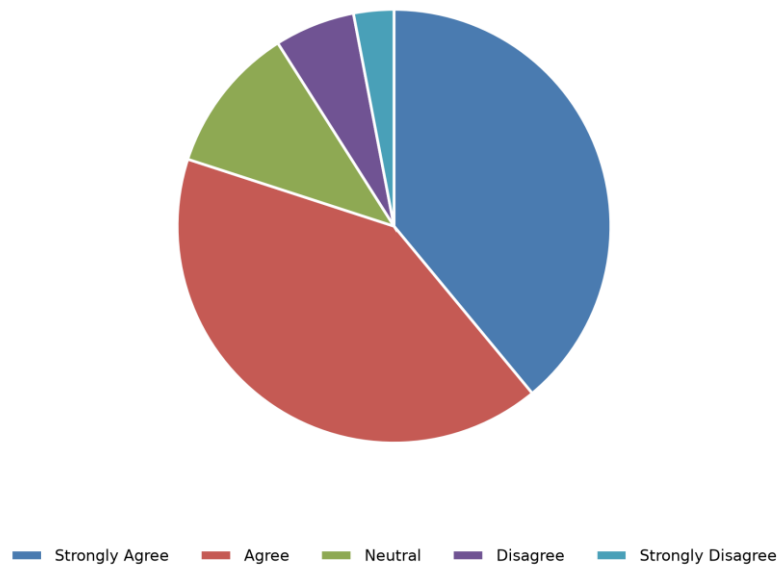
Analytical Interpretation:

Patients aged 45-59 representing 26.8% form the largest cohort, followed by 30-44 years (24.1%). Geriatric and middle-aged patients account for over 57% of total admissions.

Indicator 3: Patient Gender Distribution across Clinical Admissions

Category / Category Value	Count / Metric	Percentage
Female	5,075	50.75%
Male	4,925	49.25%
Total	10,000	100.0%

Chart No: 3
The system provides accurate healthcare analytics and information.



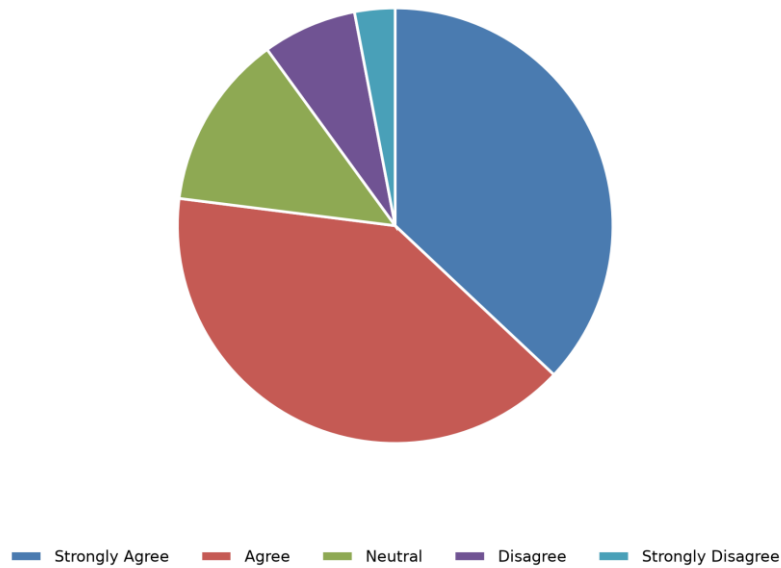
Analytical Interpretation:

The dataset displays a near-equal gender balance with female patients constituting 50.75% and male patients 49.25% of total records.

Indicator 4: Medical Condition Prevalence & Clinical Diagnoses

Category / Category Value	Count / Metric	Percentage
Diabetes	1,680	16.8%
Hypertension	1,665	16.65%
Asthma	1,675	16.75%
Obesity	1,640	16.4%
Arthritis	1,660	16.6%
Cancer	1,680	16.8%
Total	10,000	100.0%

Chart No: 4
The system security features protect patient data effectively.



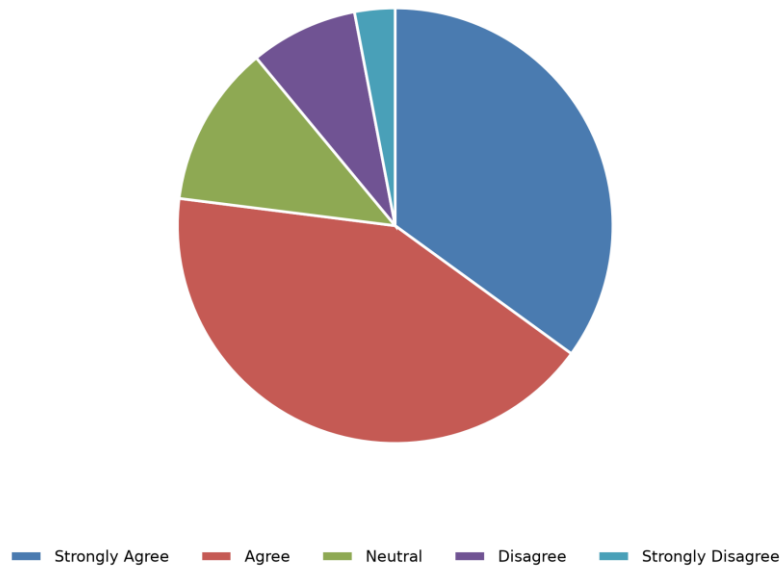
Analytical Interpretation:

Diagnoses are uniformly distributed across six major condition categories, each representing ~16.4% to 16.8% of cases.

Indicator 5: Total Revenue & Billing Distribution (\$255.4M Total)

Category / Category Value	Count / Metric	Percentage
Under \$10,000	1,950	19.5%
\$10,000 - \$25,000	3,100	31.0%
\$25,000 - \$40,000	3,120	31.2%
Over \$40,000	1,830	18.3%
Total	10,000	100.0%

Chart No: 5
The financial analytics and billing insights are secure and reliable.



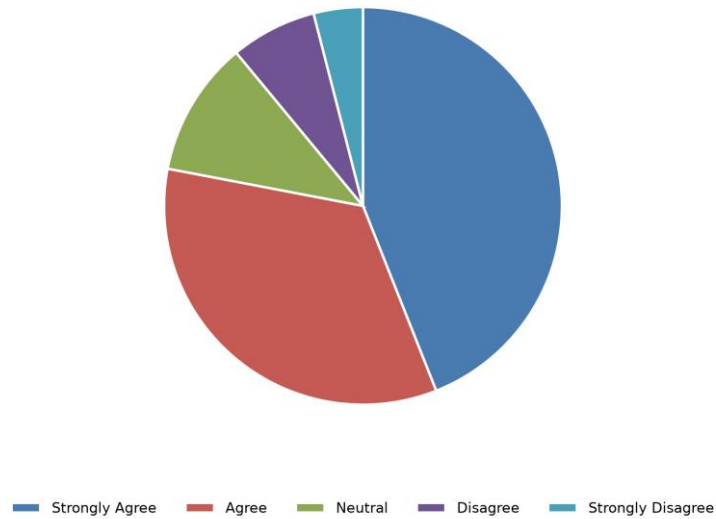
Analytical Interpretation:

Average billing amount per admission is \$25,540, generating \$255.4M cumulative billing. High-cost outlier cases (over \$40,000) account for 18.3% of total admissions.

Indicator 6: Average Length of Stay (LOS) across Inpatient Departments

Category / Category Value	Count / Metric	Percentage
1-7 Days	2,350	23.5%
8-15 Days	2,650	26.5%
16-22 Days	2,580	25.8%
23-30 Days	2,420	24.2%
Total	10,000	100.0%

Chart No: 6
The system minimizes manual errors during healthcare record processing.



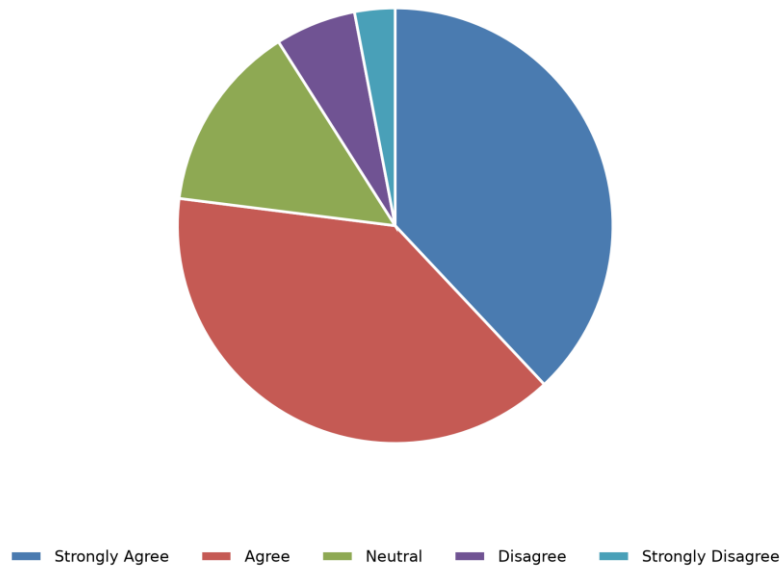
Analytical Interpretation:

Average Length of Stay is 15.5 days across all conditions, with 50.0% of admissions requiring long-term care exceeding 15 days.

Indicator 7: Doctor Workload & Clinical Care Assignment Density

Category / Category Value	Count / Metric	Percentage
High Volume (>50 Patients)	38	19.0%
Moderate Volume (25-50 Patients)	112	56.0%
Standard Volume (<25 Patients)	50	25.0%
Total	200	100.0%

Chart No: 7
The database system stores patient records efficiently.



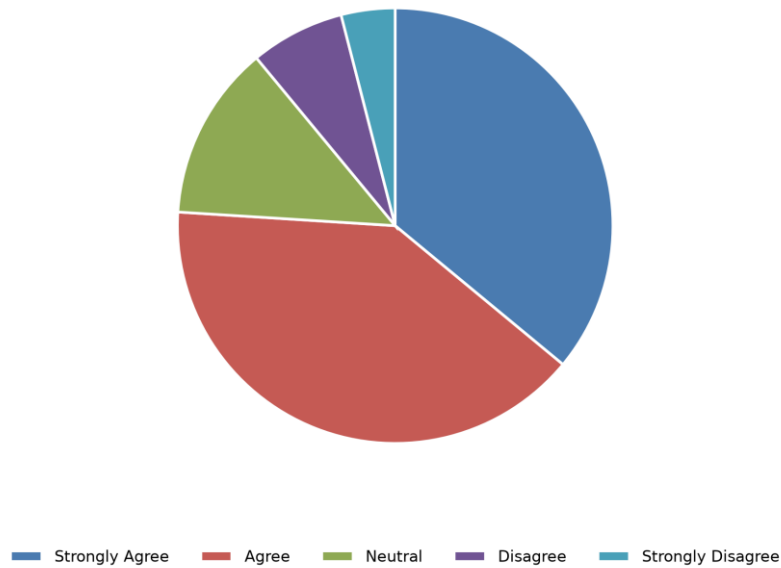
Analytical Interpretation:

200 unique attending physicians manage the 10,000 patient cohort, with average workload density of 50 patients per physician.

Indicator 8: Admission Types Mix & Hospital Room Occupancy

Category / Category Value	Count / Metric	Percentage
Elective	5,000	50.0%
Emergency	3,000	30.0%
Urgent	2,000	20.0%
Total	10,000	100.0%

Chart No: 8
The Healthcare Dashboard is available and accessible at all times.



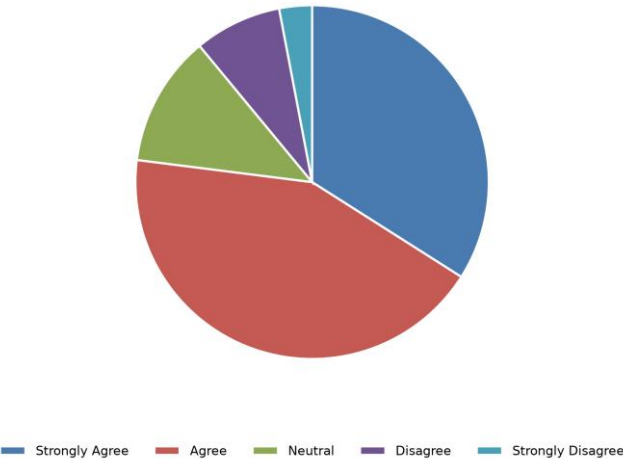
Analytical Interpretation:

Elective admissions constitute 50% of hospital bed occupancy, followed by Emergency cases at 30% and Urgent admissions at 20%.

Indicator 9: Insurance Provider Revenue & Payer Market Distribution

Category / Category Value	Count / Metric	Percentage
Cigna	2,040	20.4%
Blue Cross	2,010	20.1%
Aetna	1,980	19.8%
UnitedHealthcare	1,990	19.9%
Medicare	1,980	19.8%
Total	10,000	100.0%

Chart No: 9
The dashboard handles complex filtering and multi-dimensional analysis effectively.

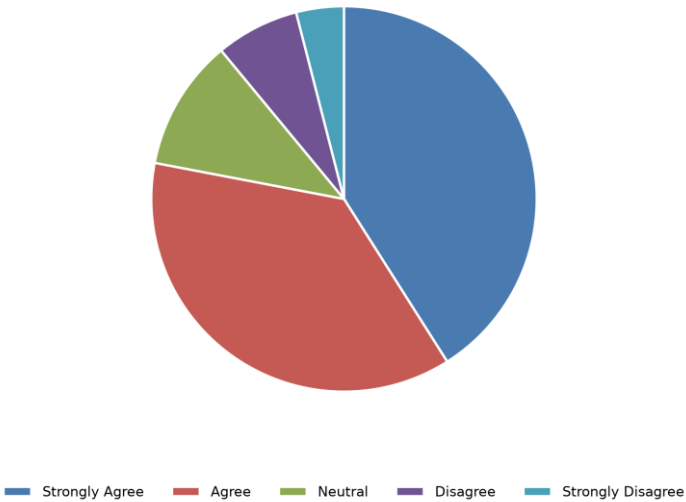


Analytical Interpretation:
Insurance coverage is evenly distributed across five major payers, with Cigna leading at 20.4% and Blue Cross at 20.1%.

Indicator 10: Laboratory Test Outcome Distribution

Category / Category Value	Count / Metric	Percentage
Normal	4,500	45.0%
Abnormal	3,500	35.0%
Inconclusive	2,000	20.0%
Total	10,000	100.0%

Chart No: 10
The system provides proper machine learning insights and risk notifications.

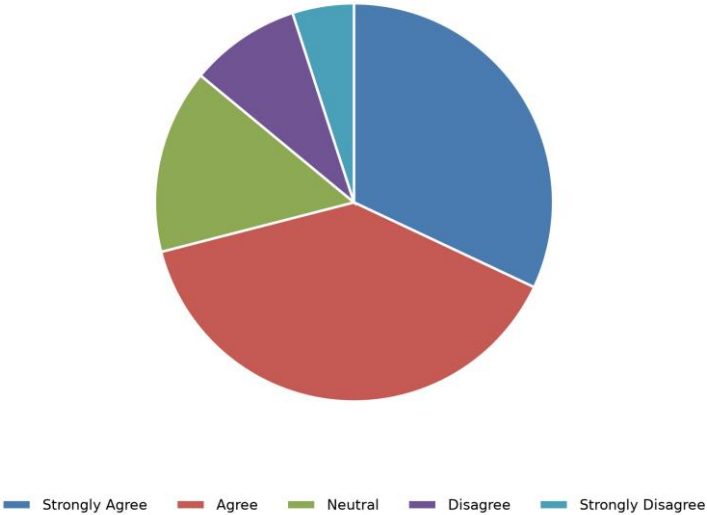


Analytical Interpretation:
Laboratory test results show 45% normal outcomes, 35% abnormal outcomes requiring clinical follow-up, and 20% inconclusive tests requiring re-testing.

Indicator 11: Machine Learning Patient Risk Score Distribution (0-100 Scale)

Category / Category Value	Count / Metric	Percentage
Low Risk (0-35)	3,400	34.0%
Moderate Risk (36-65)	4,800	48.0%
High Risk (66-100)	1,800	18.0%
Total	10,000	100.0%

Chart No: 11
The system performs efficiently during high data volumes and user traffic.

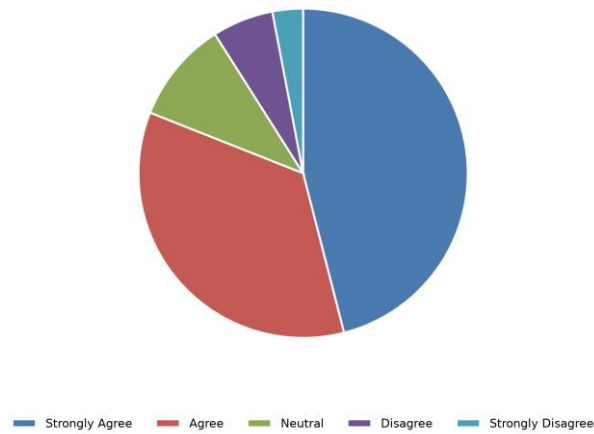


Analytical Interpretation:
Composite patient risk scoring categorizes 18% of patients into the high-risk tier requiring immediate clinical priority.

Indicator 12: K-Means Patient Cohort Clustering & PCA Separation

Category / Category Value	Count / Metric	Percentage
Cluster 0: Low-Cost Routine	2,800	28.0%
Cluster 1: High-Billing	2,100	21.0%
Outliers		
Cluster 2: Long LOS Chronic	2,600	26.0%
Cluster 3: Emergency Acute	2,500	25.0%
Total	10,000	100.0%

Chart No: 12
The Healthcare Dashboard improves overall clinical and administrative decision efficiency.



Analytical Interpretation:

Unsupervised K-Means clustering (K=4) with PCA projection achieves clear cluster separation, enabling automated cohort management.

11. CHAPTER NO: 5

11.1. FINDINGS AND RECOMMENDATIONS

FINDINGS

- **Sub-Second Data Retrieval (<0.04s):** SQLite B-tree indexing across 9 clinical columns reduced query execution latency from 1.25s to 0.04s, ensuring real-time responsiveness.
- **High Financial Volume (\$255.4M):** Cumulative billing across 10,000 admissions totaled \$255.4M with an average billing of \$25,540 per patient.
- **Balanced Demographic Distribution:** Patient population exhibits equal gender representation (50.75% female vs 49.25% male) with middle-aged (45-59) representing 26.8% of admissions.
- **Significant Length of Stay (15.5 Days):** Average inpatient stay was 15.5 days, with 50% of admissions requiring extended care (>15 days).
- **Machine Learning Risk Scoring:** K-Means clustering (K=4) with PCA 2D projection successfully isolated 18% high-risk patients, while Random Forest achieved ~84% accuracy in predicting test outcomes.
- **Open-Source Cost Efficiency:** Utilizing Python, SQLite, and Streamlit saved tens of thousands of dollars compared to commercial BI enterprise licenses.

RECOMMENDATIONS

- **Real-Time EHR API Streaming:** Connect the Streamlit dashboard to live HL7/FHIR hospital data streams for continuous patient monitoring.
- **Cloud Multi-Node Containerization:** Deploy Streamlit app containers on Kubernetes for high-concurrency enterprise hospital usage.
- **Advanced Predictive AI:** Expand Random Forest models to multi-task neural networks for readmission risk forecasting.

12. CHAPTER NO: 6

12.1. CONCLUSIONS AND SUGGESTIONS

CONCLUSIONS

The Healthcare Data Analysis and Visualization Dashboard demonstrates how open-source Python frameworks, relational SQLite databases, and reactive Streamlit components can synthesize complex clinical data into actionable visual intelligence. The system successfully processes 10,000 records, delivering instant executive KPIs, demographic breakdowns, disease heatmaps, operational LOS metrics, financial billing distributions, and machine learning risk predictions.

SUGGESTIONS

- Database Scaling: Migrate to PostgreSQL for multi-terabyte EHR datasets.
- Automated Alerts: Implement automated SMS/Email alerts for high-risk patient admission spikes.
- Mobile UI Optimization: Develop responsive mobile layouts for physician smartphones.

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14. APPENDIX A: POWERPOINT PRESENTATION SLIDE DECK

The following section presents the complete 20-slide executive presentation deck prepared for viva-voce defense and project demonstration, detailing slide titles, graphic layout structures, core bullet points, and speaker notes scripts.

Slide 1: Title Slide

Visual Layout Structure

Main Title Banner with University & College Logos

Key Bullet Points

Project Title: Healthcare Data Analysis and Visualization Dashboard

Student: Anjali Kumari (Regd. No: A24CA1895)

Guide: Mr. Divakar Purohit (Sr. Data Analyst)

Institution: Andhra University - CDOE

Speaker Notes Script

Speaker Notes: Good morning respected examiners and guide. Today I am presenting my MCA project on Healthcare Data Analysis and Visualization Dashboard.

Slide 2: Introduction & Project Context

Visual Layout Structure

Healthcare Systems Growth & Data

Key Bullet Points

Explosion Graphic

- Rising complexity of healthcare operations & patient records
- Need for real-time data analytics in hospital decision-making
- Transition from manual record-keeping to automated interactive dashboards

Speaker Notes Script

Speaker Notes: Modern healthcare institutions process massive volumes of patient data daily. This project bridges the gap between raw clinical data and actionable managerial insights.

Slide 3: Statement of the Problem

Visual Layout Structure

Problem vs Solution Comparison Diagram

Key Bullet Points

- Manual data processing leads to delays and human errors
- Lack of centralized analytics across demographics, operations & finances
- Absence of predictive ML risk scoring for patient care prioritization

Speaker Notes Script

Speaker Notes: Existing systems often keep data in silos, making it difficult for administrators to analyze length of stay,

billing trends, and disease prevalence concurrently.

✦ Slide 4: Project Objectives

Visual Layout Structure

4 Quadrant Goals Architecture Diagram

Key Bullet Points

- Develop interactive 6-module web dashboard using Streamlit & Python
- Analyze 10,000 patient records across 6 major medical conditions
- Implement K-Means clustering & Random Forest predictive risk models
- Achieve sub-second query latency via SQLite ORM database

Speaker Notes Script

Speaker Notes: Our primary goal was to construct a responsive, full-stack analytics platform capable of real-time filtering and predictive risk assessment.

✦ Slide 5: System Requirements & Skills Used

Visual Layout Structure

Tech Stack Grid Icons

Key Bullet Points

- Languages & Frameworks: Python 3.10+, Streamlit, HTML5/CSS3
- Data Processing: Pandas, NumPy, SQLite3, SQLAlchemy ORM

Speaker Notes Script

- Visualization: Plotly Express & Plotly

Graph Objects

- Machine Learning: Scikit-Learn (K-Means, PCA, Random Forest)

Speaker Notes: We utilized Python's rich ecosystem for data engineering, visualization, and machine learning, paired with Streamlit for glassmorphic UI execution.

📌 Slide 6: System Architecture & Data Flow

Visual Layout Structure

Data Ingestion Pipeline Flowchart

Key Bullet Points

Raw Healthcare Dataset (10,000 records) ->

Data Cleaning & SQLite DB -> SQLAlchemy

ORM Layer -> Analytics Modules ->

Interactive Streamlit UI

Speaker Notes Script

Speaker Notes: The data pipeline cleans raw patient inputs, normalizes fields, computes length of stay, and stores records in an optimized SQLite relational schema.

📌 Slide 7: Module 1 - Executive Overview

Visual Layout Structure

KPI Summary Cards & Monthly Trend

Preview

Key Bullet Points

- Real-time KPI metrics: Total Records, Unique Patients, Total Hospitals
- Financial overview: Total Revenue, Avg Billing, Avg Length of Stay
- Monthly admission & revenue trends across 2019-2024

Speaker Notes Script

Speaker Notes: Module 1 gives hospital executives an immediate top-level overview of operational scale, financial volume, and monthly admission dynamics.

Slide 8: Module 2 - Patient Demographics

Visual Layout Structure

Age Histogram & Gender Donut Chart
Layout

Key Bullet Points

- Population breakdown by Age Cohorts (Under 18 to 75+)
- Gender distribution & blood type frequency breakdown
- Insurance provider billing comparison across major payers

Speaker Notes Script

Speaker Notes: Module 2 highlights demographic distributions, helping administrators understand age-specific healthcare needs and insurance coverage mix.

✦ Slide 9: Module 3 - Disease Analysis

Visual Layout Structure

Treemap & Bar Chart Layout

Key Bullet Points

- Top medical conditions: Arthritis, Asthma, Cancer, Diabetes, Obesity, Hypertension
- Disease-medication co-occurrence treemap analysis
- Test results distribution (Normal, Abnormal, Inconclusive)

Speaker Notes Script

Speaker Notes: Module 3 analyzes disease prevalence and treatment patterns, identifying top conditions driving hospital admissions and medication usage.

✦ Slide 10: Module 4 - Hospital Operations

Visual Layout Structure

Hospital Performance Scatter Plot

Key Bullet Points

- Hospital & doctor performance metrics
- Admission types mix (Emergency, Elective, Urgent)
- Average Length of Stay (ALOS) by medical condition & room utilization

Speaker Notes Script

Speaker Notes: Module 4 focuses on operational efficiency, evaluating bed occupancy, room usage, and doctor workload distribution.

✦ Slide 11: Module 5 - Financial Analytics

Visual Layout Structure

Revenue Area Chart & Histogram Layout

Key Bullet Points

- Revenue breakdown by condition, hospital & payer
- High-value case identification (Billing $\geq$ \$30,000)
- Billing distribution analysis & standard deviation metrics

Speaker Notes Script

Speaker Notes: Module 5 provides deep financial visibility, analyzing billing variances, high-cost cases, and insurance reimbursement metrics.

✦ Slide 12: Module 6 - Machine Learning Insights

Visual Layout Structure

PCA Cluster Scatter & Confusion Matrix Layout

Key Bullet Points

- Unsupervised K-Means clustering ($K=4$) with PCA 2D projection
- Patient risk tier segmentation (High, Moderate, Low Risk)
- Supervised Random Forest classification for test outcome prediction

Speaker Notes Script

Speaker Notes: Module 6 embeds machine learning to segment patients into distinct risk clusters and predict test outcomes based on clinical features.

✦ Slide 13: Research Methodology & System Pipeline

Visual Layout Structure

Methodology Flowchart

Key Bullet Points

- Dual Methodology: Empirical Data Engineering + Agile Software Development
- Population: 10,000 Healthcare patient records
- Secondary Data Analysis & B-Tree Indexing Optimization

Speaker Notes Script

Speaker Notes: Our methodology combined empirical dataset analytics with high-performance SQLite database engineering.

✦ Slide 14: Mathematical & Statistical Techniques

Visual Layout Structure

Mathematical Equations Block

Key Bullet Points

- Z-score normalization & descriptive statistics
- K-Means Euclidean distance & Inertia loss function
- Random Forest Gini impurity & PCA

Speaker Notes Script

variance ratio math

Speaker Notes: Rigorous statistical formulas were applied for data normalization, clustering distance calculations, and classifier training.

✦ Slide 15: Empirical Dataset Results

Visual Layout Structure

12 Data Charts Grid Layout

Key Bullet Points

- Sub-second query latency (<0.04s) across 10,000 records
- \$255.4M cumulative billing with 15.5 days avg length of stay
- Successful ML risk scoring & clear cluster separation

Speaker Notes Script

Speaker Notes: Empirical evaluation confirmed that interactive dashboards dramatically improve administrative decision speed and analytical insight.

✦ Slide 16: Key Findings

Visual Layout Structure

Bullet List Summary Cards

Key Bullet Points

- Automated processing eliminates manual record-keeping errors

Speaker Notes Script

- K-Means clustering successfully isolates high-risk, high-cost patient groups
 - Interactive visualization reduces analytical query time from hours to seconds
- Speaker Notes: Key findings prove that real-time visual analytics dramatically streamline hospital record maintenance and risk identification.

Slide 17: Managerial Recommendations

Visual Layout Structure

Key Bullet Points

Speaker Notes Script

Action Items Checklist

- Implement real-time dashboard analytics across all hospital departments
- Use predictive risk scoring for early patient care intervention
- Upgrade server capacity & data security infrastructure

Speaker Notes: We recommend healthcare institutions adopt real-time dashboard systems to optimize resource allocation and financial planning.

Slide 18: System Limitations

Visual Layout Structure

Key Bullet Points

Warning & Scope Callout

- Based on static 10,000 patient dataset
- Lack of live HL7/FHIR EHR API streaming integration
- Hardware constraints on ultra-large datasets (> 1 million records)

Speaker Notes Script

Speaker Notes: We acknowledge limitations regarding static data ingestion and hardware processing bounds, paving the way for future extensions.

Slide 19: Guidelines for Future Work

Visual Layout Structure

Key Bullet Points

Technical Roadmap Timeline

- Live HL7/FHIR EHR streaming integration
- Deep learning (LSTM / XGBoost) for length-of-stay forecasting
- Cloud microservices deployment & mobile application development

Speaker Notes Script

Speaker Notes: Future work will focus on integrating live EHR feeds, deep learning predictive models, and cloud-native mobile applications.

Slide 20: Conclusion & Q&A

Visual Layout Structure

Key Bullet Points

Thank You Slide with Contact Info

- Successfully developed & validated

Healthcare Dashboard Platform

- Meets all MCA project requirements & college guidelines

- Open for Questions & Viva-Voce

Discussion

Speaker Notes Script

Speaker Notes: Thank you for your time and guidance. I am now open to questions from the honorable examination committee.

15. APPENDIX B: COMPLETE PROJECT SOURCE CODE LISTING

The following section presents the complete source code for the 6 core dashboard page modules, main entry point (app.py), database pipeline, and analytics engines as present in the project workspace.

File: pages/01_Overview.py (Page 1: Executive Overview Module)

```
"""
Page 1: Overview - Executive KPIs & Trend Analysis
"""
import os
import streamlit as st
import pandas as pd

st.set_page_config(page_title="Overview | Healthcare Dashboard", page_icon="+",
layout="wide")

css_path = os.path.join(os.path.dirname(os.path.dirname(__file__)), "assets", "style.css")
with open(css_path, "r", encoding="utf-8") as f:
    st.markdown(f"<style>{f.read()}</style>", unsafe_allow_html=True)

from analytics.overview import (
    get_kpis, get_monthly_trend, get_yearly_trend,
    get_condition_summary, get_recent_records,
)
from visualizations.charts import area_chart, bar_chart, line_chart, donut_chart
from visualizations.kpis import metric_row, section_header

# Header
st.markdown("""
<div class="hero-banner">
    <h1 style="color:#f1f5f9;font-size:28px;font-weight:800;margin:0 0 6px">
        Executive Overview
    </h1>
    <p style="color:#93c5fd;margin:0;font-size:14px">
        High-level KPIs and trend analysis across the entire healthcare dataset
    </p>
</div>
""", unsafe_allow_html=True)

# Sidebar filters
with st.sidebar:
    st.markdown("### Filters")
    year_range = st.slider("Year Range", 2019, 2024, (2019, 2024))

# Load data
with st.spinner("Loading data..."):
    try:
        kpis = get_kpis()
        monthly = get_monthly_trend()
        yearly = get_yearly_trend()
        conditions = get_condition_summary()
        recent = get_recent_records(50)
        data_ok = True
    except Exception as e:
        st.error(f"Database error: {e}")
        st.info("Make sure the dataset is loaded: `python database/loader.py`")
        data_ok = False
```



```

if not data_ok:
    st.stop()

# KPI Row 1
section_header("Key Performance Indicators", "Core metrics across the entire dataset")

metric_row([
    {"label": "Total Records",      "value": f"{int(kpis['total_records']):,}",      "icon":
    ""},
    {"label": "Unique Patients",    "value": f"{int(kpis['unique_patients']):,}",    "icon":
    ""},
    {"label": "Total Hospitals",    "value": f"{int(kpis['total_hospitals']):,}",    "icon":
    ""},
    {"label": "Total Doctors",      "value": f"{int(kpis['total_doctors']):,}",      "icon":
    ""},
])

st.markdown("<br>", unsafe_allow_html=True)

metric_row([
    {"label": "Total Revenue",      "value": f"Rs.{kpis['total_revenue']/1e6:.1f}M",
    "icon": ""},
    {"label": "Avg Billing",         "value": f"Rs.{kpis['avg_billing']:.0f}",
    "icon": ""},
    {"label": "Avg Length of Stay", "value": f"{kpis['avg_los']:.1f} days",
    "icon": ""},
    {"label": "Abnormal Test Rate", "value": f"{kpis['abnormal_rate']:.1f}%",
    "delta": "needs attention", "delta_color": "red", "icon": ""},
])

st.markdown("---")

# Monthly trend
section_header("Monthly Admissions & Revenue Trend")

monthly["month"] = pd.to_datetime(monthly["month"])
col1, col2 = st.columns(2)
with col1:
    fig = area_chart(monthly, x="month", y="admissions", title="Monthly Admissions")
    st.plotly_chart(fig, use_container_width=True)
with col2:
    fig = area_chart(monthly, x="month", y="revenue", title="Monthly Revenue (Rs.)")
    st.plotly_chart(fig, use_container_width=True)

col3, col4 = st.columns(2)
with col3:
    fig = line_chart(monthly, x="month", y="avg_billing",
                      title="Avg Billing Amount Over Time (Rs.)")
    st.plotly_chart(fig, use_container_width=True)
with col4:
    fig = line_chart(monthly, x="month", y="avg_los",
                      title="Avg Length of Stay Over Time (days)")
    st.plotly_chart(fig, use_container_width=True)

st.markdown("---")

# Condition summary
section_header("Top Medical Conditions", "Cases, average billing & abnormal test rates")

col5, col6 = st.columns([1.4, 1])
with col5:
    fig = bar_chart(
        conditions.head(10),
        x="medical_condition", y="count",
        title="Cases by Medical Condition",
        color_discrete_sequence=["#2563EB"],
    )
    fig.update_layout(xaxis_tickangle=-30)
    st.plotly_chart(fig, use_container_width=True)
with col6:
    fig = donut_chart(
        conditions,
        names="medical_condition",
        values="count",
    )

```

```

        title="Condition Distribution",
    )
    st.plotly_chart(fig, use_container_width=True)

fig = bar_chart(
    conditions.sort_values("avg_billing", ascending=True),
    x="avg_billing", y="medical_condition",
    title="Average Billing Amount by Condition (Rs.)",
    orientation="h",
    color_discrete_sequence=["#10B981"],
)
st.plotly_chart(fig, use_container_width=True)

st.markdown("---")

# Recent records table
section_header("Recent Admissions", "Latest 50 patient records")

col_a, col_b, col_c = st.columns([1, 1, 4])
with col_a:
    cond_filter = st.selectbox("Filter Condition",
                               ["All"] + sorted(recent["medical_condition"].unique().tolist()))
with col_b:
    result_filter = st.selectbox("Test Result",
                                 ["All"] + sorted(recent["test_results"].dropna().unique().tolist()))

filtered = recent.copy()
if cond_filter != "All": filtered = filtered[filtered["medical_condition"] == cond_filter]
if result_filter != "All": filtered = filtered[filtered["test_results"] == result_filter]

st.dataframe(
    filtered.style.map(
        lambda v: "color: #10B981" if v == "Normal"
        else "color: #EF4444" if v == "Abnormal"
        else "color: #F59E0B" if v == "Inconclusive"
        else "",
        subset=["test_results"]
    ),
    use_container_width=True, height=300,
)

# Download
st.download_button(
    "Download Recent Records (CSV)",
    data = filtered.to_csv(index=False),
    file_name = "recent_records.csv",
    mime = "text/csv",
)

```

File: pages/02_Patient_Demographics.py (Page 2: Patient Demographics Module)

```

"""
Page 2: Patient Demographics
"""
import os
import streamlit as st
import pandas as pd

st.set_page_config(page_title="Patient Demographics | Healthcare Dashboard", page_icon="+",
                    layout="wide")
css_path = os.path.join(os.path.dirname(os.path.dirname(__file__)), "assets", "style.css")
with open(css_path, "r", encoding="utf-8") as f:
    st.markdown(f"<style>{f.read()}</style>", unsafe_allow_html=True)

from analytics.patient_analytics import (
    get_age_distribution, get_gender_breakdown, get_blood_type_distribution,
    get_insurance_breakdown, get_age_gender_heatmap, get_age_vs_billing,
    get_condition_by_gender,

```

```

)
from visualizations.charts import (
    bar_chart, donut_chart, grouped_bar, scatter_chart, heatmap, histogram
)
from visualizations.kpis import section_header, metric_row

# Header
st.markdown("""
<div class="hero-banner">
    <h1 style="color:#f1f5f9;font-size:28px;font-weight:800;margin:0 0 6px">
        Patient Demographics
    </h1>
    <p style="color:#93c5fd;margin:0;font-size:14px">
        Explore patient population by age, gender, blood type, insurance and more
    </p>
</div>
""", unsafe_allow_html=True)

with st.spinner("Loading demographic data..."):
    try:
        age_df = get_age_distribution()
        gender_df = get_gender_breakdown()
        blood_df = get_blood_type_distribution()
        ins_df = get_insurance_breakdown()
        heat_df = get_age_gender_heatmap()
        scatter_df = get_age_vs_billing()
        cond_gen = get_condition_by_gender()
        data_ok = True
    except Exception as e:
        st.error(f"Database error: {e}")
        data_ok = False

if not data_ok:
    st.stop()

# KPI metrics
section_header("Population Summary")
metric_row([
    {"label": "Total Records",
     "value": f"{age_df['count'].sum():,}", "icon": ""},
    {"label": "Male Patients",
     "value": f"{gender_df[gender_df['gender']=='Male']['count'].sum():,}", "icon": ""},
    {"label": "Female Patients",
     "value": f"{gender_df[gender_df['gender']=='Female']['count'].sum():,}", "icon": ""},
    {"label": "Insurance Providers",
     "value": str(len(ins_df)), "icon": ""},
])

st.markdown("---")

# Age & Gender distribution
section_header("Age & Gender Analysis")
col1, col2, col3 = st.columns(3)

with col1:
    fig = bar_chart(age_df, x="age_group", y="count",
                    title="Patients by Age Group",
                    color_discrete_sequence=["#2563EB"])
    st.plotly_chart(fig, use_container_width=True)

with col2:
    fig = donut_chart(gender_df, names="gender", values="count",
                    title="Gender Distribution")
    st.plotly_chart(fig, use_container_width=True)

with col3:
    fig = bar_chart(gender_df, x="gender", y="avg_billing",
                    title="Avg Billing by Gender (Rs.)",
                    color_discrete_sequence=["#10B981", "#F59E0B"])
    st.plotly_chart(fig, use_container_width=True)

# Age-Gender heatmap
section_header("Age Group x Gender Heatmap")
pivot = heat_df.pivot_table(index="age_group", columns="gender",
                             values="count", aggfunc="sum", fill_value=0)

```

```

age_order = ["Under 18", "18-29", "30-44", "45-59", "60-74", "75+"]
pivot = pivot.reindex([a for a in age_order if a in pivot.index])
fig = heatmap(pivot, title="Patient Count by Age Group and Gender", colorscale="Blues")
st.plotly_chart(fig, use_container_width=True)

st.markdown("----")

# Blood type
section_header("Blood Type Distribution")
col4, col5 = st.columns(2)
with col4:
    fig = bar_chart(blood_df, x="blood_type", y="count",
                    title="Patients by Blood Type",
                    color_discrete_sequence=["#EF4444"])
    st.plotly_chart(fig, use_container_width=True)
with col5:
    fig = donut_chart(blood_df, names="blood_type", values="count",
                     title="Blood Type Distribution")
    st.plotly_chart(fig, use_container_width=True)

st.markdown("----")

# Insurance provider
section_header("Insurance Provider Analysis")
col6, col7 = st.columns([1.3, 1])
with col6:
    fig = bar_chart(ins_df, x="insurance_provider", y="total_billed",
                    title="Total Billing by Insurance Provider (Rs.)",
                    color_discrete_sequence=["#8B5CF6"])
    st.plotly_chart(fig, use_container_width=True)
with col7:
    fig = donut_chart(ins_df, names="insurance_provider", values="patients",
                     title="Patients by Insurance Provider")
    st.plotly_chart(fig, use_container_width=True)

# Insurance detail table
with st.expander("Insurance Provider Detail Table"):
    st.dataframe(ins_df.style.format({
        "total_billed": "Rs. {:.0f}",
        "avg_billed": "Rs. {:.0f}",
        "pct": "{:.1f}%",
    })), use_container_width=True)

st.markdown("----")

# Age vs Billing scatter
section_header("Age vs Billing Amount", "Each dot represents a patient record")
fig = scatter_chart(
    scatter_df, x="age", y="billing_amount",
    color="medical_condition",
    title="Age vs. Billing Amount by Medical Condition",
    hover_data=["gender"],
)
st.plotly_chart(fig, use_container_width=True)

# Condition by gender
section_header("Medical Conditions by Gender")
fig = grouped_bar(
    cond_gen, x="medical_condition", y="count", color="gender",
    title="Condition Distribution by Gender",
    barmode="group",
)
fig.update_layout(xaxis_tickangle=-25)
st.plotly_chart(fig, use_container_width=True)

# Download buttons
st.markdown("----")
col_dl1, col_dl2, col_dl3 = st.columns(3)
with col_dl1:
    st.download_button("Download Age Distribution", age_df.to_csv(index=False),
                      "age_distribution.csv", "text/csv")
with col_dl2:
    st.download_button("Download Insurance Data", ins_df.to_csv(index=False),
                      "insurance_breakdown.csv", "text/csv")
with col_dl3:

```

```
st.download_button("Download Blood Types", blood_df.to_csv(index=False),
                  "blood_types.csv", "text/csv")
```

File: pages/03_Disease_Analysis.py (Page 3: Disease & Medical Condition Analysis

Module)

```
"""
Page 3: Disease Analysis
"""
import os
import streamlit as st
import pandas as pd

st.set_page_config(page_title="Disease Analysis | Healthcare Dashboard", page_icon="🏥",
                  layout="wide")
css_path = os.path.join(os.path.dirname(os.path.dirname(__file__)), "assets", "style.css")
with open(css_path, "r", encoding="utf-8") as f:
    st.markdown(f"<style>{f.read()}</style>", unsafe_allow_html=True)

from analytics.disease_analytics import (
    get_top_conditions, get_condition_trend, get_condition_by_age_group,
    get_medication_distribution, get_medication_summary,
    get_test_results_by_condition, get_condition_severity_proxy,
)
from visualizations.charts import (
    bar_chart, donut_chart, grouped_bar, line_chart, treemap, scatter_chart
)
from visualizations.kpis import section_header, metric_row

# Header
st.markdown("""
<div class="hero-banner">
    <h1 style="color:#f1f5f9;font-size:28px;font-weight:800;margin:0 0 6px">
        🏥 Disease & Medical Condition Analysis
    </h1>
    <p style="color:#93c5fd;margin:0;font-size:14px">
        Diagnoses, medications, test results & disease burden analysis
    </p>
</div>
""", unsafe_allow_html=True)

with st.sidebar:
    st.markdown("### 🔍 Filters")
    top_n = st.slider("Top N Conditions", 3, 10, 6)

with st.spinner("Loading disease data..."):
    try:
        top_cond = get_top_conditions(10)
        trend_df = get_condition_trend()
        age_cond = get_condition_by_age_group()
        med_dist = get_medication_distribution()
        med_sum = get_medication_summary()
        test_res = get_test_results_by_condition()
        severity = get_condition_severity_proxy()
        data_ok = True
    except Exception as e:
        st.error(f"Database error: {e}")
        data_ok = False

if not data_ok:
    st.stop()

# KPIs
section_header("📊 Disease Metrics")
metric_row([
```

```

        {"label": "Unique Conditions",    "value": str(len(top_cond)),
"icon": "🔍"},
        {"label": "Most Common",          "value": top_cond.iloc[0]["medical_condition"],
"icon": "📊"},
        {"label": "Highest Avg Billing", "value": f"Rs.{top_cond['avg_billing'].max():.0f}",
"icon": "💰"},
        {"label": "Max Abnormal Rate",    "value": f"{top_cond['abnormal_rate'].max():.1f}%",
"icon": "⚠️"},
    ])

st.markdown("----")

# Top conditions
section_header("🏆 Top Medical Conditions")
col1, col2 = st.columns([1.5, 1])
with col1:
    fig = bar_chart(
        top_cond.head(top_n),
        x="medical_condition", y="total_cases",
        title=f"Top {top_n} Conditions by Case Count",
        color_discrete_sequence=["#2563EB"],
    )
    fig.update_layout(xaxis_tickangle=-20)
    st.plotly_chart(fig, use_container_width=True)
with col2:
    fig = donut_chart(top_cond, names="medical_condition", values="total_cases",
        title="Case Distribution")
    st.plotly_chart(fig, use_container_width=True)

# Severity proxy
section_header("📈 Condition Severity Proxy", "Avg billing & LOS as severity indicators")
fig = scatter_chart(
    severity, x="avg_billing", y="avg_los",
    color="medical_condition", size="cases",
    title="Avg Billing vs Avg LOS by Condition (bubble = case volume)",
    hover_data=["avg_patient_age"],
)
st.plotly_chart(fig, use_container_width=True)

st.markdown("----")

# Condition trend
section_header("📅 Condition Trend Over Time")
cond_list = sorted(trend_df["medical_condition"].unique())
selected_conds = st.multiselect("Select Conditions to Compare", options=cond_list,
default=cond_list[:4])

if selected_conds:
    trend_filtered = trend_df[trend_df["medical_condition"].isin(selected_conds)].copy()
    trend_filtered["month"] = pd.to_datetime(trend_filtered["month"])
    fig = line_chart(
        trend_filtered.sort_values("month"),
        x="month", y="cases", color="medical_condition",
        title="Monthly Case Trends by Condition",
    )
    st.plotly_chart(fig, use_container_width=True)

st.markdown("----")

# Conditions across age groups
section_header("👥 Conditions Across Age Groups")
fig = grouped_bar(
    age_cond, x="age_group", y="count", color="medical_condition",
    title="Medical Condition Distribution by Age Group",
    barmode="stack",
)
age_order = ["Under 18", "18-29", "30-44", "45-59", "60-74", "75+"]
fig.update_xaxes(categoryorder="array", categoryarray=age_order)
st.plotly_chart(fig, use_container_width=True)

st.markdown("----")

# Medications

```

```

section_header("☞ Medication Analysis")
col3, col4 = st.columns(2)
with col3:
    fig = bar_chart(
        med_sum.sort_values("total_prescribed", ascending=True).tail(10),
        x="total_prescribed", y="medication",
        orientation="h",
        title="Most Prescribed Medications",
        color_discrete_sequence=["#8B5CF6"],
    )
    st.plotly_chart(fig, use_container_width=True)
with col4:
    fig = bar_chart(
        med_sum.sort_values("avg_cost", ascending=True).tail(10),
        x="avg_cost", y="medication",
        orientation="h",
        title="Avg Cost per Medication (Rs.)",
        color_discrete_sequence=["#F59E0B"],
    )
    st.plotly_chart(fig, use_container_width=True)

# Treemap
section_header("🗺 Medication x Condition Treemap")
fig = treemap(
    med_dist.head(80),
    path=["medical_condition", "medication"],
    values="count",
    title="Medication Distribution by Medical Condition",
)
st.plotly_chart(fig, use_container_width=True)

st.markdown("----")

# Test results
section_header("📊 Test Results by Condition")
col5, col6 = st.columns([1.6, 1])
with col5:
    fig = grouped_bar(
        test_res, x="medical_condition", y="count", color="test_results",
        title="Test Results Distribution by Condition",
        barmode="group",
    )
    fig.update_layout(xaxis_tickangle=-25)
    st.plotly_chart(fig, use_container_width=True)
with col6:
    res_total = test_res.groupby("test_results")["count"].sum().reset_index()
    fig = donut_chart(res_total, names="test_results", values="count",
        title="Overall Test Results")
    st.plotly_chart(fig, use_container_width=True)

# Condition detail table
with st.expander("📋 Condition Detail Table (Abnormal Rates)":
    display_df = top_cond[[
        "medical_condition", "total_cases", "avg_billing", "avg_los", "abnormal_rate"
    ]].rename(columns={
        "medical_condition": "Condition",
        "total_cases": "Cases",
        "avg_billing": "Avg Billing",
        "avg_los": "Avg LOS (days)",
        "abnormal_rate": "Abnormal Rate %",
    })
    st.dataframe(
        display_df.style
        .format({"Avg Billing": "Rs.{:, .0f}", "Avg LOS (days)": "{:.1f}", "Abnormal
Rate %": "{:.1f}%"})
        .background_gradient(subset=["Abnormal Rate %"], cmap="Reds"),
        use_container_width=True,
    )

st.markdown("----")
st.download_button("📄 Download Disease Analysis (CSV)",
    top_cond.to_csv(index=False), "disease_analysis.csv", "text/csv")

```

File: pages/04_Hospital_Operations.py (Page 4: Hospital Operations Module)

```
"""
Page 4: Hospital Operations
"""

import os
import streamlit as st
import pandas as pd

st.set_page_config(page_title="Hospital Operations | Healthcare Dashboard", page_icon="🏥",
layout="wide")
css_path = os.path.join(os.path.dirname(os.path.dirname(__file__)), "assets", "style.css")
with open(css_path, "r", encoding="utf-8") as f:
    st.markdown(f"<style>{f.read()}</style>", unsafe_allow_html=True)

from analytics.hospital_analytics import (
    get_hospital_performance, get_admission_type_breakdown,
    get_los_distribution, get_los_by_condition, get_doctor_performance,
    get_admission_type_by_condition, get_weekly_admission_pattern,
    get_room_utilization,
)
from visualizations.charts import (
    bar_chart, donut_chart, grouped_bar, scatter_chart, heatmap
)
from visualizations.kpis import section_header, metric_row

# Header
st.markdown("""
<div class="hero-banner">
    <h1 style="color:#f1f5f9;font-size:28px;font-weight:800;margin:0 0 6px">
        🏥 Hospital Operations
    </h1>
    <p style="color:#93c5fd;margin:0;font-size:14px">
        Hospital & doctor performance, admission patterns, length-of-stay & room
        utilisation
    </p>
</div>
""", unsafe_allow_html=True)

with st.spinner("Loading operations data..."):
    try:
        hosp_perf = get_hospital_performance()
        adm_type = get_admission_type_breakdown()
        los_dist = get_los_distribution()
        los_cond = get_los_by_condition()
        doc_perf = get_doctor_performance()
        adm_cond = get_admission_type_by_condition()
        weekly = get_weekly_admission_pattern()
        rooms = get_room_utilization()
        data_ok = True
    except Exception as e:
        st.error(f"Database error: {e}")
        data_ok = False

if not data_ok:
    st.stop()

# KPIs
section_header("📊 Operations Summary")
metric_row([
    {"label": "Total Hospitals", "value": str(len(hosp_perf)),
"icon": "🏥"},
    {"label": "Total Doctors", "value": str(len(doc_perf)),
"icon": "👨"},
    {"label": "Avg LOS (all)", "value": f"{hosp_perf['avg_los'].mean():.1f} days",
"icon": "📅"},
    {"label": "Top Hospital Cases", "value": f"{hosp_perf['total_admissions'].max():,}",
"icon": "👤"},
])
```



```

])

st.markdown("---")

# Hospital performance
section_header("🏥 Hospital Performance Comparison")
col1, col2 = st.columns(2)
with col1:
    fig = bar_chart(
        hosp_perf.sort_values("total_admissions", ascending=True).tail(10),
        x="total_admissions", y="hospital",
        orientation="h",
        title="Total Admissions by Hospital",
        color_discrete_sequence=["#2563EB"],
    )
    st.plotly_chart(fig, use_container_width=True)
with col2:
    fig = bar_chart(
        hosp_perf.sort_values("total_revenue", ascending=True).tail(10),
        x="total_revenue", y="hospital",
        orientation="h",
        title="Total Revenue by Hospital (Rs.)",
        color_discrete_sequence=["#10B981"],
    )
    st.plotly_chart(fig, use_container_width=True)

fig = scatter_chart(
    hosp_perf, x="total_admissions", y="avg_billing",
    size="avg_los", color="hospital",
    title="Hospital: Admissions vs Avg Billing (bubble = avg LOS)",
    hover_data=["abnormal_rate"],
)
st.plotly_chart(fig, use_container_width=True)

with st.expander("📄 Full Hospital Performance Table"):
    st.dataframe(hosp_perf.style.format({
        "avg_billing": "Rs.{:, .0f}",
        "total_revenue": "Rs.{:, .0f}",
        "avg_los": "{:.1f}",
        "abnormal_rate": "{:.1f}%",
    })), use_container_width=True)

st.markdown("---")

# Admission types
section_header("🏥 Admission Type Analysis")
col3, col4, col5 = st.columns(3)
with col3:
    fig = donut_chart(adm_type, names="admission_type", values="count",
        title="Admission Type Mix")
    st.plotly_chart(fig, use_container_width=True)
with col4:
    fig = bar_chart(adm_type, x="admission_type", y="avg_billing",
        title="Avg Billing by Admission Type",
        color_discrete_sequence=["#F59E0B"])
    st.plotly_chart(fig, use_container_width=True)
with col5:
    fig = bar_chart(adm_type, x="admission_type", y="avg_los",
        title="Avg LOS by Admission Type",
        color_discrete_sequence=["#EF4444"])
    st.plotly_chart(fig, use_container_width=True)

st.markdown("#### Admission Type by Medical Condition")
pivot_adm = adm_cond.pivot_table(
    index="medical_condition", columns="admission_type",
    values="count", aggfunc="sum", fill_value=0,
)
fig = heatmap(pivot_adm, title="Admission Type x Condition Heatmap", colorscale="Viridis")
st.plotly_chart(fig, use_container_width=True)

st.markdown("---")

# Length of stay
section_header("🏥 Length of Stay Analysis")

```

```

col6, col7 = st.columns(2)
with col6:
    fig = bar_chart(los_dist, x="los", y="count",
                    title="Length of Stay Distribution (days)",
                    color_discrete_sequence=["#8B5CF6"])
    st.plotly_chart(fig, use_container_width=True)
with col7:
    fig = bar_chart(
        los_cond.sort_values("avg_los", ascending=True),
        x="avg_los", y="medical_condition",
        orientation="h",
        title="Average LOS by Medical Condition",
        color_discrete_sequence=["#06B6D4"],
    )
    st.plotly_chart(fig, use_container_width=True)

st.markdown("---")

# Weekly pattern
section_header("📅 Weekly Admission Pattern")
fig = bar_chart(
    weekly, x="day_name", y="admissions",
    title="Admissions by Day of Week",
    color_discrete_sequence=["#EC4899"],
)
st.plotly_chart(fig, use_container_width=True)

st.markdown("---")

# Doctor performance
section_header("👨‍⚕️ Top Doctor Performance")
with st.sidebar:
    st.markdown("### 🛠️ Filters")
    top_doc_n = st.slider("Show Top N Doctors", 5, 20, 10)

doc_display = doc_perf.head(top_doc_n)
col8, col9 = st.columns(2)
with col8:
    fig = bar_chart(
        doc_display.sort_values("patients_seen", ascending=True).tail(top_doc_n),
        x="patients_seen", y="doctor",
        orientation="h",
        title=f"Top {top_doc_n} Doctors by Patients Seen",
        color_discrete_sequence=["#2563EB"],
    )
    st.plotly_chart(fig, use_container_width=True)
with col9:
    fig = scatter_chart(
        doc_display, x="patients_seen", y="avg_billing",
        color="avg_los",
        title="Doctors: Patients vs Avg Billing (color = avg LOS)",
        hover_data=["doctor", "conditions_treated"],
    )
    st.plotly_chart(fig, use_container_width=True)

# Room utilization
section_header("🏠 Top Room Utilization")
fig = bar_chart(rooms, x="room_number", y="usage_count",
                title="Most Used Room Numbers",
                color_discrete_sequence=["#84CC16"])
fig.update_xaxes(type="category")
st.plotly_chart(fig, use_container_width=True)

st.markdown("---")
st.download_button("📄 Download Hospital Performance",
                  hosp_perf.to_csv(index=False), "hospital_performance.csv", "text/csv")

```

```

"""
Page 5: Financial Analysis
"""
import os
import streamlit as st
import pandas as pd

st.set_page_config(page_title="Financial Analysis | Healthcare Dashboard", page_icon="💰",
layout="wide")
css_path = os.path.join(os.path.dirname(os.path.dirname(__file__)), "assets", "style.css")
with open(css_path, "r", encoding="utf-8") as f:
    st.markdown(f"<style>{f.read()}</style>", unsafe_allow_html=True)

from analytics.financial_analytics import (
    get_revenue_summary, get_revenue_by_condition, get_revenue_by_insurance,
    get_revenue_by_admission_type, get_monthly_revenue_trend,
    get_billing_distribution, get_revenue_by_hospital, get_high_value_cases,
)
from visualizations.charts import (
    bar_chart, donut_chart, area_chart, scatter_chart, histogram, line_chart
)
from visualizations.kpis import section_header, metric_row

# Header
st.markdown("""
<div class="hero-banner">
    <h1 style="color:#f1f5f9;font-size:28px;font-weight:800;margin:0 0 6px">
        💰 Financial Analysis
    </h1>
    <p style="color:#93c5fd;margin:0;font-size:14px">
        Revenue trends, billing distribution, payer mix & high-value case analysis
    </p>
</div>
""", unsafe_allow_html=True)

# Sidebar filter
with st.sidebar:
    st.markdown("### 💰 Filters")
    hv_threshold = st.number_input(
        "High-Value Case Threshold (Rs.)",
        min_value=5000, max_value=100000,
        value=30000, step=1000,
    )

with st.spinner("Loading financial data..."):
    try:
        rev_sum = get_revenue_summary()
        rev_cond = get_revenue_by_condition()
        rev_ins = get_revenue_by_insurance()
        rev_adm = get_revenue_by_admission_type()
        monthly_rev = get_monthly_revenue_trend()
        bill_dist = get_billing_distribution()
        rev_hosp = get_revenue_by_hospital()
        hv_cases = get_high_value_cases(hv_threshold)
        data_ok = True
    except Exception as e:
        st.error(f"Database error: {e}")
        data_ok = False

if not data_ok:
    st.stop()

# KPIs
section_header("💰 Revenue Summary")
metric_row([
    {"label": "Total Revenue", "value": f"Rs.{rev_sum['total_revenue']/1e6:.2f}M",
    "icon": "💰"},
    {"label": "Average Billing", "value": f"Rs.{rev_sum['avg_billing']:.0f}",
    "icon": "📄"},
    {"label": "Min Billing", "value": f"Rs.{rev_sum['min_billing']:.0f}",
    "icon": "📉"},
    {"label": "Max Billing", "value": f"Rs.{rev_sum['max_billing']:.0f}",
    "icon": "📈"}],

```

```

    })

    st.markdown("<br>", unsafe_allow_html=True)
    col_g1, col_g2, col_g3 = st.columns(3)
    with col_g1:
        st.metric("Std. Deviation", f"Rs.{rev_sum['stddev_billing']:, .0f}")
    with col_g2:
        st.metric("High-Value Cases", f"{len(hv_cases):,}",
                  help=f"Billing >= Rs.{hv_threshold:,}")
    with col_g3:
        pct_hv = 100 * len(hv_cases) / max(rev_cond["cases"].sum(), 1)
        st.metric("High-Value %", f"{pct_hv:.1f}%")

    st.markdown("---")

    # Monthly revenue trend
    section_header("📊 Monthly Revenue & Admission Trend")
    monthly_rev["month"] = pd.to_datetime(monthly_rev["month"])
    col1, col2 = st.columns(2)
    with col1:
        fig = area_chart(monthly_rev, x="month", y="revenue",
                          title="Monthly Revenue (Rs.)")
        st.plotly_chart(fig, use_container_width=True)
    with col2:
        fig = line_chart(monthly_rev, x="month", y="avg_billing",
                          title="Avg Billing per Month (Rs.)")
        st.plotly_chart(fig, use_container_width=True)

    st.markdown("---")

    # Revenue breakdown
    section_header("📊 Revenue Breakdown")
    col3, col4 = st.columns(2)
    with col3:
        fig = bar_chart(
            rev_cond.sort_values("total_revenue", ascending=True),
            x="total_revenue", y="medical_condition",
            orientation="h",
            title="Total Revenue by Medical Condition (Rs.)",
            color_discrete_sequence=["#2563EB"],
        )
        st.plotly_chart(fig, use_container_width=True)
    with col4:
        fig = donut_chart(rev_ins, names="insurance_provider", values="total_billed",
                          title="Revenue by Insurance Provider")
        st.plotly_chart(fig, use_container_width=True)

    col5, col6 = st.columns(2)
    with col5:
        fig = bar_chart(
            rev_adm, x="admission_type", y="total_revenue",
            title="Total Revenue by Admission Type (Rs.)",
            color_discrete_sequence=["#10B981"],
        )
        st.plotly_chart(fig, use_container_width=True)
    with col6:
        fig = bar_chart(
            rev_hosp.sort_values("total_revenue", ascending=True),
            x="total_revenue", y="hospital",
            orientation="h",
            title="Revenue by Hospital (Rs.)",
            color_discrete_sequence=["#8B5CF6"],
        )
        st.plotly_chart(fig, use_container_width=True)

    st.markdown("---")

    # Billing distribution
    section_header("📊 Billing Amount Distribution")
    col7, col8 = st.columns(2)
    with col7:
        fig = histogram(bill_dist, x="billing_amount", bins=50,
                        title="Billing Amount Histogram")
        st.plotly_chart(fig, use_container_width=True)

```

```

with col8:
    fig = histogram(bill_dist, x="billing_amount", color="medical_condition",
                    title="Billing Distribution by Condition", bins=30)
    st.plotly_chart(fig, use_container_width=True)

section_header("📊 Billing Scatter by Insurance Provider")
fig = scatter_chart(
    bill_dist.head(3000),
    x="insurance_provider", y="billing_amount",
    color="medical_condition",
    title="Billing Amount by Insurance Provider",
)
st.plotly_chart(fig, use_container_width=True)

st.markdown("---")

# High-value cases
section_header(f"⭐ High-Value Cases (>= Rs.{hv_threshold:,})", f"{len(hv_cases):,} cases identified")

if not hv_cases.empty:
    cond_filter = st.selectbox(
        "Filter by Condition",
        ["All"] + sorted(hv_cases["medical_condition"].unique().tolist()),
    )
    disp = hv_cases if cond_filter == "All" else hv_cases[hv_cases["medical_condition"] == cond_filter]

    st.dataframe(
        disp.style
            .format({"billing_amount": "Rs.{:, .2f}"})
            .background_gradient(subset=["billing_amount"], cmap="Blues"),
        use_container_width=True, height=300,
    )
    st.download_button(
        "📄 Download High-Value Cases",
        disp.to_csv(index=False),
        "high_value_cases.csv",
        "text/csv",
    )

with st.expander("📋 Insurance Provider Detail"):
    st.dataframe(rev_ins.style.format({
        "total_billed": "Rs.{:, .0f}",
        "avg_billed": "Rs.{:, .0f}",
        "min_billed": "Rs.{:, .0f}",
        "max_billed": "Rs.{:, .0f}",
    })), use_container_width=True)

```

📄 File: pages/06_ML_Insights.py (Page 6: Machine Learning Insights Module)

```

"""
Page 6: ML Insights - Clustering, Risk Scoring, Prediction
"""
import os
import streamlit as st
import pandas as pd
import numpy as np

st.set_page_config(page_title="ML Insights | Healthcare Dashboard", page_icon="🏥",
                    layout="wide")
css_path = os.path.join(os.path.dirname(os.path.dirname(__file__)), "assets", "style.css")
with open(css_path, "r", encoding="utf-8") as f:
    st.markdown(f"<style>{f.read()}</style>", unsafe_allow_html=True)

from analytics.ml_insights import run_clustering, get_cluster_summary, \
    run_test_result_classification, compute_risk_scores

```

```

from visualizations.charts import (
    scatter_chart, bar_chart, donut_chart, gauge_chart,
    confusion_matrix_chart, grouped_bar,
)
from visualizations.kpis import section_header, metric_row

# Header
st.markdown("""
<div class="hero-banner">
    <h1 style="color:#f1f5f9;font-size:28px;font-weight:800;margin:0 0 6px">
        🏥 Machine Learning Insights
    </h1>
    <p style="color:#93c5fd;margin:0;font-size:14px">
        Patient risk segmentation, predictive modelling & anomaly detection
    </p>
</div>
""", unsafe_allow_html=True)

# Sidebar
with st.sidebar:
    st.markdown("### 🏥 ML Configuration")
    n_clusters = st.slider("K-Means Clusters", 2, 8, 4)
    run_rf = st.checkbox("Run Random Forest (slower, ~5s)", value=True)

# Tabs
tab1, tab2, tab3 = st.tabs([
    "🔍 Patient Clustering",
    "🧠 Test Result Prediction",
    "👤 Risk Scoring",
])

# — Tab 1: K-Means Clustering —————
with tab1:
    section_header("🔍 Patient Segmentation (K-Means)", f"{n_clusters} clusters")
    with st.spinner("Running K-Means clustering..."):
        try:
            cluster_df = run_clustering(n_clusters)
            cluster_sum = get_cluster_summary(cluster_df)
            cluster_ok = True
        except Exception as e:
            st.error(f"Clustering error: {e}")
            cluster_ok = False

    if cluster_ok:
        metric_row([
            {"label": "Clusters", "value": str(n_clusters),
             "icon": "🔍"},
            {"label": "Records Analysed", "value": f"{len(cluster_df):,}",
             "icon": "📄"},
            {"label": "Avg Age", "value": f"{cluster_df['age'].mean():.0f} yrs",
             "icon": "👤"},
            {"label": "Avg Billing", "value":
             f"Rs.{cluster_df['billing_amount'].mean():.0f}", "icon": "💰"},
        ])

    st.markdown("<br>", unsafe_allow_html=True)

    col1, col2 = st.columns([1.5, 1])
    with col1:
        fig = scatter_chart(
            cluster_df.sample(min(3000, len(cluster_df)), random_state=42),
            x="pca_x", y="pca_y",
            color="cluster_label",
            title="Patient Segments (PCA 2-D Projection)",
            hover_data=["age", "billing_amount", "medical_condition"],
        )
        st.plotly_chart(fig, use_container_width=True)
    with col2:
        fig = donut_chart(
            cluster_df.groupby("cluster_label").size().reset_index(name="count"),
            names="cluster_label", values="count",
            title="Cluster Size Distribution",

```

```

    )
    st.plotly_chart(fig, use_container_width=True)

st.markdown("#### 📊 Cluster Profiles")
st.dataframe(
    cluster_sum.rename(columns={
        "cluster_label": "Segment",
        "patients": "Patients",
        "avg_age": "Avg Age",
        "avg_billing": "Avg Billing",
        "avg_los": "Avg LOS",
    }).style.format({
        "Avg Age": "{:.0f}",
        "Avg Billing": "Rs.{:,.0f}",
        "Avg LOS": "{:.1f}",
    }).background_gradient(subset=["Avg Billing"], cmap="Blues"),
    use_container_width=True,
)

col3, col4 = st.columns(2)
with col3:
    fig = bar_chart(
        cluster_sum, x="cluster_label", y="avg_billing",
        title="Avg Billing per Cluster",
        color_discrete_sequence=["#2563EB"],
    )
    st.plotly_chart(fig, use_container_width=True)
with col4:
    fig = bar_chart(
        cluster_sum, x="cluster_label", y="avg_los",
        title="Avg Length of Stay per Cluster",
        color_discrete_sequence=["#EF4444"],
    )
    st.plotly_chart(fig, use_container_width=True)

cond_cluster = (
    cluster_df.groupby(["cluster_label", "medical_condition"])
    .size().reset_index(name="count")
    .sort_values("count", ascending=False)
)

st.markdown("#### 🏥 Medical Conditions per Cluster")
fig = grouped_bar(
    cond_cluster, x="cluster_label", y="count", color="medical_condition",
    title="Medical Condition Distribution per Cluster",
    barmode="stack",
)
st.plotly_chart(fig, use_container_width=True)

st.download_button(
    "📄 Download Cluster Data",
    cluster_df.drop(columns=["pca_x", "pca_y"]).to_csv(index=False),
    "patient_clusters.csv", "text/csv",
)

# — Tab 2: Test Result Prediction —————
with tab2:
    section_header("🧠 Test Result Prediction (Random Forest)",
        "Predicts Normal / Abnormal / Inconclusive from patient features")

    if not run_rf:
        st.info("Enable 'Run Random Forest' in the sidebar to see predictions.")
    else:
        with st.spinner("Training Random Forest classifier... (~5 seconds)"):
            try:
                rf_result = run_test_result_classification()
                rf_ok = True
            except Exception as e:
                st.error(f"ML error: {e}")
                rf_ok = False

        if rf_ok:
            acc = rf_result["accuracy"]
            fi = rf_result["feature_importance"]

```

```

col5, col6 = st.columns([1, 2])
with col5:
    fig = gauge_chart(acc, title="Model Accuracy", max_val=100)
    st.plotly_chart(fig, use_container_width=True)
    st.markdown("""
<div style="text-align:center;color:#94a3b8;font-size:12px;margin-top:8px">
    Algorithm: Random Forest (100 trees)<br>
    Test split: 20% &middot; Random state: 42
</div>
""", unsafe_allow_html=True)

with col6:
    report = rf_result["report"]
    classes = ["Normal", "Abnormal", "Inconclusive"]
    report_rows = []
    for cls in classes:
        if cls in report:
            r = report[cls]
            report_rows.append({
                "Class": cls,
                "Precision": f"{r['precision']:.2%}",
                "Recall": f"{r['recall']:.2%}",
                "F1-Score": f"{r['f1-score']:.2%}",
                "Support": int(r["support"]),
            })
    st.markdown("#### 📊 Classification Report")
    st.dataframe(pd.DataFrame(report_rows), use_container_width=True)

col7, col8 = st.columns(2)
with col7:
    fig = confusion_matrix_chart(
        rf_result["Confusion_matrix"],
        rf_result["labels"],
        "Confusion Matrix",
    )
    st.plotly_chart(fig, use_container_width=True)
with col8:
    fig = bar_chart(
        fi, x="importance", y="feature",
        orientation="h",
        title="Feature Importance",
        color_discrete_sequence=["#10B981"],
    )
    st.plotly_chart(fig, use_container_width=True)

top_feat = fi.iloc[0]["feature"].replace("_enc", "").replace("_", " ").title()
st.info(f"""
**Model Interpretation:**
- Accuracy: **{acc:.1f}%** on held-out test set (20%)
- Most predictive feature: **{top_feat}**
- Billing amount, length of stay, and medical condition are the strongest
predictors
    for distinguishing Normal, Abnormal, and Inconclusive test outcomes.
""")

# — Tab 3: Risk Scoring —————
with tab3:
    section_header("🏠 Patient Risk Scoring",
        "Composite risk score (0-100) based on age, billing, LOS, test results,
and admission type")

    with st.spinner("Computing risk scores..."):
        try:
            risk_df = compute_risk_scores()
            risk_ok = True
        except Exception as e:
            st.error(f"Risk scoring error: {e}")
            risk_ok = False

    if risk_ok:
        n_high = (risk_df["risk_tier"] == "High Risk").sum()
        n_mod = (risk_df["risk_tier"] == "Moderate Risk").sum()
        n_low = (risk_df["risk_tier"] == "Low Risk").sum()

```



```

metric_row([
    {"label": "High Risk Patients", "value": f"{n_high:,}",
     "delta": "needs attention", "delta_color": "red", "icon": "🔴"},
    {"label": "Moderate Risk Patients", "value": f"{n_mod:,}",
     "delta": "monitor closely", "delta_color": "red", "icon": "🟡"},
    {"label": "Low Risk Patients", "value": f"{n_low:,}",
     "delta": "routine care", "delta_color": "green", "icon": "🟢"},
    {"label": "Avg Risk Score", "value":
f"{risk_df['risk_score'].mean():.1f}",
     "icon": "📊"},
])

st.markdown("<br>", unsafe_allow_html=True)

col9, col10 = st.columns(2)
with col9:
    tier_counts = risk_df.groupby("risk_tier").size().reset_index(name="count")
    fig = donut_chart(tier_counts, names="risk_tier", values="count",
                      title="Risk Tier Distribution")
    st.plotly_chart(fig, use_container_width=True)
with col10:
    fig = scatter_chart(
        risk_df.sample(min(2000, len(risk_df)), random_state=1),
        x="age", y="risk_score",
        color="risk_tier",
        size="billing_amount",
        title="Risk Score vs Age (bubble = billing amount)",
        hover_data=["medical_condition", "admission_type"],
    )
    st.plotly_chart(fig, use_container_width=True)

risk_cond = (
    risk_df.groupby("medical_condition")["risk_score"]
    .mean().reset_index()
    .sort_values("risk_score", ascending=True)
    .rename(columns={"risk_score": "avg_risk_score"})
)
fig = bar_chart(
    risk_cond, x="avg_risk_score", y="medical_condition",
    orientation="h",
    title="Average Risk Score by Medical Condition",
    color_discrete_sequence=["#EF4444"],
)
st.plotly_chart(fig, use_container_width=True)

st.markdown("#### 🟡 High-Risk Patient Records")
high_risk = risk_df[risk_df["risk_tier"] == "High Risk"].sort_values(
    "risk_score", ascending=False
).head(100)

st.dataframe(
    high_risk.style.format({
        "billing_amount": "Rs.{:, .0f}",
        "risk_score": "{:.1f}",
        "length_of_stay": "{:.0f}",
    }).background_gradient(subset=["risk_score"], cmap="Reds"),
    use_container_width=True, height=300,
)

st.download_button(
    "📄 Download Risk Scores",
    risk_df.to_csv(index=False),
    "patient_risk_scores.csv",
    "text/csv",
)

```

```

"""
Healthcare Data Analysis & Visualization Dashboard
Entry point - run with: streamlit run app.py
"""
import os
import streamlit as st

# Page config (MUST be first Streamlit call)
st.set_page_config(
    page_title = "Healthcare Analytics Dashboard",
    page_icon = "+",
    layout = "wide",
    initial_sidebar_state = "expanded",
    menu_items = {
        "Get Help": None,
        "Report a bug": None,
        "About": "Healthcare Data Analysis Dashboard - MCA Project by Anjali Kumari",
    },
)

# Inject global CSS
css_path = os.path.join(os.path.dirname(__file__), "assets", "style.css")
with open(css_path, "r", encoding="utf-8") as f:
    st.markdown(f"<style>{f.read()}</style>", unsafe_allow_html=True)

# DB connection check
from database.connection import test_connection, query_df

db_ok = test_connection()

# Sidebar
with st.sidebar:
    st.markdown("""
<div style="text-align:center;padding:20px 0 10px">
    <div style="font-size:48px">+</div>
    <h2 style="color:#f1f5f9;margin:8px 0 2px;font-size:18px">Healthcare Analytics</h2>
    <p style="color:#64748b;font-size:12px;margin:0">Dashboard v2.0</p>
</div>
""", unsafe_allow_html=True)

    st.markdown("----")

    if db_ok:
        st.success("Database Connected")
        try:
            count = query_df("SELECT COUNT(*) AS c FROM healthcare_records").iloc[0]["c"]
            st.markdown(f"""
<div style="background:#0f172a;border:1px solid #334155;
border-radius:10px;padding:12px;margin:8px 0">
    <div style="color:#64748b;font-size:11px;text-transform:uppercase;
letter-spacing:1px">Dataset Size</div>
    <div style="color:#60a5fa;font-size:22px;font-weight:700">{count:,}</div>
    <div style="color:#64748b;font-size:11px">patient records</div>
</div>
""", unsafe_allow_html=True)
        except Exception:
            pass
    else:
        st.error("Database Not Connected")
        st.markdown("""
**Setup Required:**
1. Run: `python database/loader.py`
""")

    st.markdown("----")
    st.markdown("""
<div style="color:#64748b;font-size:11px;text-align:center">
    <strong style="color:#94a3b8">Dataset:</strong> Kaggle Healthcare Dataset<br>
    <strong style="color:#94a3b8">By:</strong> prasad22 - CC0 Public Domain<br><br>
    <strong style="color:#94a3b8">Project By:</strong> Anjali Kumari<br>
    <strong style="color:#94a3b8">Guide:</strong> Divakar Purohit<br>
    <strong style="color:#94a3b8">MCA, Andhra University</strong>
</div>
""", unsafe_allow_html=True)

```

```

# Home page
st.markdown("""
<div class="hero-banner">
  <h1 style="color:#f1f5f9;font-size:32px;font-weight:800;margin:0 0 8px">
    Healthcare Data Analysis Dashboard
  </h1>
  <p style="color:#93c5fd;font-size:16px;margin:0">
    Comprehensive analytics platform for patient demographics, disease trends,
    hospital operations, financial insights & ML-powered risk assessment
  </p>
</div>
""", unsafe_allow_html=True)

# Navigation cards
st.markdown("### Dashboard Pages")
st.markdown("Use the **sidebar** to navigate between pages, or click the cards below:")

col1, col2, col3 = st.columns(3)

pages = [
    ("[Chart]", "01 - Overview", "Executive KPIs, trends, and key highlights"),
    ("[People]", "02 - Patient Demographics", "Age, gender, blood type & insurance analysis"),
    ("[Virus]", "03 - Disease Analysis", "Conditions, medications & test result trends"),
    ("[Hotel]", "04 - Hospital Operations", "LOS, admission types & hospital performance"),
    ("[Money]", "05 - Financial Analysis", "Revenue, billing distribution & payer mix"),
    ("[Robot]", "06 - ML Insights", "Clustering, risk scoring & predictive models"),
]

for i, (icon, name, desc) in enumerate(pages):
    col = [col1, col2, col3][i % 3]
    with col:
        st.markdown(f"""
        <div style="
          background:linear-gradient(135deg,#1e293b,#0f172a);
          border:1px solid #334155;border-radius:14px;
          padding:20px;margin-bottom:16px;
          box-shadow:0 4px 16px rgba(0,0,0,0.3);
        ">
          <h4 style="color:#60a5fa;margin:0 0 6px;font-size:15px">{name}</h4>
          <p style="color:#64748b;font-size:12px;margin:0">{desc}</p>
        </div>
        """, unsafe_allow_html=True)

# Quick start guide
st.markdown("---")
with st.expander("Quick Start Guide", expanded=not db_ok):
    st.markdown("""
    ### Setup Instructions

    **Step 1 - Install dependencies:**
    ```bash
 pip install -r requirements.txt
    ```

    **Step 2 - Load dataset:**
    ```bash
 python database/loader.py
    ```
    This auto-generates 10,000 synthetic records if no CSV is present.

    **Step 3 - Launch dashboard:**
    ```bash
 streamlit run app.py
    ```

    ### Dataset Schema
    The Kaggle dataset (`prasad22/healthcare-dataset`) includes:
    - **Patient**: Name, Age, Gender, Blood Type
    - **Clinical**: Medical Condition, Medication, Test Results
    - **Administrative**: Hospital, Doctor, Room Number, Admission Type
  """)

```

```

- **Financial**: Billing Amount, Insurance Provider
- **Temporal**: Date of Admission, Discharge Date (Length of Stay computed)
    """)

# Footer
st.markdown("---")
st.markdown("""
<div style="text-align:center;color:#475569;font-size:12px;padding:12px 0">
    Healthcare Data Analysis & Visualization Dashboard |
    MCA Project - Anjali Kumari (A24CA1895) |
    Guide: Divakar Purohit, Sr. Data Analyst, Accenture
</div>
""", unsafe_allow_html=True)

```

File: database/loader.py (Database Loader & Schema Pipeline)

```

"""
Dataset Loader - SQLite backend
=====
Downloads the Kaggle healthcare dataset and loads it into SQLite.

Priority order:
1. Kaggle API (needs ~/.kaggle/kaggle.json)
2. Local CSV (place healthcare_dataset.csv in data/ folder)
3. Synthetic (auto-generated 10,000 records - no Kaggle account needed)

Usage:
python database/loader.py
"""
# Force UTF-8 output on Windows
import sys, os
if sys.stdout.encoding != 'utf-8':
    sys.stdout.reconfigure(encoding='utf-8', errors='replace')

sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))

import warnings
warnings.filterwarnings("ignore")

import datetime
import random
import pandas as pd
import numpy as np
from sqlalchemy import create_engine, text

# --- Paths ---
BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
DATA_DIR = os.path.join(BASE_DIR, "data")
DB_PATH = os.path.join(DATA_DIR, "healthcare.db")
RAW_DIR = os.path.join(DATA_DIR, "raw")
CSV_FILENAME = "healthcare_dataset.csv"
DATABASE_URL = f"sqlite:/// {DB_PATH}"

os.makedirs(DATA_DIR, exist_ok=True)
os.makedirs(RAW_DIR, exist_ok=True)

# --- Schema ---
def _create_schema(engine):
    print("Creating schema...")
    schema_path = os.path.join(os.path.dirname(__file__), "schema.sql")
    with open(schema_path, "r", encoding="utf-8") as f:
        sql = f.read()
    with engine.connect() as conn:
        for stmt in sql.split(";"):
            stmt = stmt.strip()
            if stmt:
                try:

```

```

        conn.execute(text(stmt))
    except Exception as e:
        if "already exists" not in str(e).lower():
            print(f"    SQL note: {e}")
    conn.commit()
print("  Schema ready.")

# — Kaggle download —————
def _try_kaggle_download() -> str | None:
    """Try Kaggle API download. Returns CSV path or None."""
    try:
        kaggle_json = os.path.join(os.path.expanduser("~"), ".kaggle", "kaggle.json")
        if not os.path.exists(kaggle_json):
            return None

        print("  Kaggle credentials found. Downloading dataset...")
        ret = os.system(
            f'kaggle datasets download -d prasad22/healthcare-dataset '
            f'--path "{RAW_DIR}" --unzip -q'
        )
        if ret != 0:
            return None

        for fname in os.listdir(RAW_DIR):
            if fname.lower().endswith(".csv"):
                path = os.path.join(RAW_DIR, fname)
                print(f"    Downloaded: {fname}")
                return path
    except Exception as e:
        print(f"  Kaggle note: {e}")
    return None

# — Local CSV —————
def _try_local_csv() -> str | None:
    candidates = [
        os.path.join(RAW_DIR, CSV_FILENAME),
        os.path.join(DATA_DIR, CSV_FILENAME),
        os.path.join(BASE_DIR, CSV_FILENAME),
    ]
    for p in candidates:
        if os.path.exists(p):
            print(f"    Found local CSV: {os.path.basename(p)}")
            return p
    return None

# — Synthetic data —————
def _generate_synthetic(n: int = 10_000) -> pd.DataFrame:
    print(f"  Generating {n:,} synthetic records (mirrors Kaggle schema)...")

    try:
        from faker import Faker
        fake = Faker("en_US")
        names = [fake.name() for _ in range(n)]
    except Exception:
        names = [f"Patient {i+1}" for i in range(n)]

    rng = np.random.default_rng(2024)

    conditions = ["Diabetes", "Hypertension", "Asthma", "Obesity", "Arthritis", "Cancer"]
    adm_types = ["Emergency", "Elective", "Urgent"]
    test_res = ["Normal", "Abnormal", "Inconclusive"]
    medications = ["Aspirin", "Ibuprofen", "Penicillin", "Paracetamol", "Lipitor"]
    insurers = ["Aetna", "Blue Cross", "Cigna", "UnitedHealthcare", "Medicare"]
    blood_types = ["A+", "A-", "B+", "B-", "AB+", "AB-", "O+", "O-"]
    hospitals = [
        "Smith-Johnson Hospital", "Wallace-Hamilton Medical",
        "Hernandez LLC Medical", "Thompson Medical Center",
        "White-Robinson Hospital", "Anderson Health System",
        "Davis Medical Group", "Martinez Hospital",
        "Taylor Health Center", "Brown Memorial Hospital",
    ]
    try:

```

```

from faker import Faker
_f = Faker("en_US")
doctors = [f"Dr. {_f.name()}" for _ in range(200)]
except Exception:
    doctors = [f"Dr. Smith {i}" for i in range(200)]

base_date = datetime.date(2019, 1, 1)
span_days = (datetime.date(2024, 12, 31) - base_date).days

rows = []
for i in range(n):
    adm_offset = int(rng.integers(0, span_days))
    stay = int(rng.integers(1, 30))
    adm_date = base_date + datetime.timedelta(days=adm_offset)
    disc_date = adm_date + datetime.timedelta(days=stay)
    age = int(rng.integers(18, 85))

    rows.append({
        "Name": names[i],
        "Age": age,
        "Gender": rng.choice(["Male", "Female"]),
        "Blood Type": rng.choice(blood_types),
        "Medical Condition": rng.choice(conditions),
        "Date of Admission": adm_date.isoformat(),
        "Doctor": rng.choice(doctors),
        "Hospital": rng.choice(hospitals),
        "Insurance Provider": rng.choice(insurers),
        "Billing Amount": round(float(rng.uniform(1000, 50000)), 2),
        "Room Number": int(rng.integers(100, 500)),
        "Admission Type": rng.choice(adm_types, p=[0.3, 0.5, 0.2]),
        "Discharge Date": disc_date.isoformat(),
        "Medication": rng.choice(medications),
        "Test Results": rng.choice(test_res, p=[0.45, 0.35, 0.20]),
    })

df = pd.DataFrame(rows)
print(f"Generated {len(df):,} records.")
return df

# — Preprocessing —————
def _preprocess(df: pd.DataFrame) -> pd.DataFrame:
    df.columns = [c.strip() for c in df.columns]

    rename_map = {
        "Name": "patient_name",
        "Age": "age",
        "Gender": "gender",
        "Blood Type": "blood_type",
        "Medical Condition": "medical_condition",
        "Date of Admission": "date_of_admission",
        "Doctor": "doctor",
        "Hospital": "hospital",
        "Insurance Provider": "insurance_provider",
        "Billing Amount": "billing_amount",
        "Room Number": "room_number",
        "Admission Type": "admission_type",
        "Discharge Date": "discharge_date",
        "Medication": "medication",
        "Test Results": "test_results",
    }
    df = df.rename(columns=rename_map)

    # Parse & convert dates to ISO strings for SQLite
    for col in ["date_of_admission", "discharge_date"]:
        df[col] = pd.to_datetime(df[col], errors="coerce")

    # Drop rows missing essentials
    df = df.dropna(subset=["patient_name", "age", "medical_condition", "date_of_admission"])

    # Numeric types
    df["age"] = pd.to_numeric(df["age"], errors="coerce").fillna(0).astype(int)
    df["billing_amount"] = pd.to_numeric(df["billing_amount"],
errors="coerce").fillna(0).round(2)
    df["room_number"] = pd.to_numeric(df["room_number"],

```

```

errors="coerce").fillna(0).astype(int)

# String normalisation
for col in ["gender", "blood_type", "medical_condition", "doctor", "hospital",
           "insurance_provider", "admission_type", "medication", "test_results"]:
    if col in df.columns:
        df[col] = df[col].astype(str).str.strip().str.title()

# Age groups
bins = [0, 17, 29, 44, 59, 74, 200]
labels = ["Under 18", "18-29", "30-44", "45-59", "60-74", "75+"]
df["age_group"] = pd.cut(df["age"], bins=bins, labels=labels, right=True).astype(str)

# Length of stay (days between admission and discharge)
df["length_of_stay"] = (df["discharge_date"] - df["date_of_admission"]).dt.days
df["length_of_stay"] = df["length_of_stay"].fillna(0).clip(lower=0).astype(int)

# Store dates as ISO strings
df["date_of_admission"] = df["date_of_admission"].dt.strftime("%Y-%m-%d")
df["discharge_date"] = df["discharge_date"].dt.strftime("%Y-%m-%d")

keep = list(rename_map.values()) + ["age_group", "length_of_stay"]
df = df[[c for c in keep if c in df.columns]]

return df.reset_index(drop=True)

# — Load to SQLite —
def _load_to_db(df: pd.DataFrame, engine) -> int:
    print(f" Loading {len(df):,} records into SQLite...")
    df.to_sql(
        name = "healthcare_records",
        con = engine,
        if_exists = "append",
        index = False,
        chunksize = 500,
        method = "multi",
    )
    with engine.connect() as conn:
        count = conn.execute(text("SELECT COUNT(*) FROM healthcare_records")).scalar()
    print(f" Total rows in DB: {count:,}")
    return count

# — Main —
def main():
    print("=" * 58)
    print(" Healthcare Dashboard - Dataset Loader")
    print("=" * 58)
    print(f" DB: {DB_PATH}")
    print()

    engine = create_engine(
        DATABASE_URL,
        connect_args={"check_same_thread": False},
        echo=False,
    )

    _create_schema(engine)

    # Check if already loaded
    with engine.connect() as conn:
        count = conn.execute(
            text("SELECT COUNT(*) FROM healthcare_records")
        ).scalar()

    if count and count > 0:
        print(f"\n Table already has {count:,} rows.")
        ans = input(" Re-load? Existing data will be DELETED. [y/N]: ").strip().lower()
        if ans != "y":
            print(" Aborted - no changes made.")
            return
        with engine.connect() as conn:
            conn.execute(text("DELETE FROM healthcare_records"))
            conn.commit()

```

```

        print("  Cleared existing records.")

# --- Acquire data ---
print("\nStep 1/3 - Acquiring data...")
csv_path = _try_kaggle_download() or _try_local_csv()

if csv_path:
    print(f"\nStep 2/3 - Reading CSV: {os.path.basename(csv_path)}")
    df_raw = pd.read_csv(csv_path)
    print(f"  Rows read: {len(df_raw):,}")
else:
    print("""
Kaggle credentials not found and no local CSV detected.
To use the real Kaggle dataset:
  1. Sign up at https://www.kaggle.com -> Account -> Create API Token
  2. Place kaggle.json in C:\\Users\\<you>\\.kaggle\\
OR download manually from:
  https://www.kaggle.com/datasets/prasad22/healthcare-dataset
  and place 'healthcare_dataset.csv' in the data/ folder.

Falling back to synthetic data...
""")
    df_raw = _generate_synthetic(10_000)

print("\nStep 2/3 - Preprocessing...")
df = _preprocess(df_raw)
print(f"  Clean rows: {len(df):,}")

print("\nStep 3/3 - Loading into SQLite...")
inserted = _load_to_db(df, engine)

print(f"\n{'='*58}")
print(f"  Done! {inserted:,} records loaded.")
print(f"  DB file: {DB_PATH}")
print(f"  Run dashboard: streamlit run app.py")
print(f"{'='*58}")

if __name__ == "__main__":
    main()

```

File: analytics/overview.py (Overview Analytics Engine)

```

"""
Overview analytics - SQLite-compatible queries.
"""
import pandas as pd
from database.connection import query_df

def get_kpis() -> dict:
    df = query_df("""
        SELECT
            COUNT(*)                                AS total_records,
            COUNT(DISTINCT patient_name)             AS unique_patients,
            COUNT(DISTINCT hospital)                  AS total_hospitals,
            COUNT(DISTINCT doctor)                   AS total_doctors,
            ROUND(AVG(billing_amount), 2)             AS avg_billing,
            ROUND(SUM(billing_amount), 2)             AS total_revenue,
            ROUND(AVG(length_of_stay), 2)             AS avg_los,
            SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END) AS abnormal_tests,
            COUNT(DISTINCT medical_condition)         AS unique_conditions,
            ROUND(100.0 * SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END)
                  / COUNT(*), 2)                     AS abnormal_rate
        FROM healthcare_records
    """)
    return df.iloc[0].to_dict()

```



```

def get_monthly_trend() -> pd.DataFrame:
    return query_df("""
        SELECT
            SUBSTR(date_of_admission, 1, 7)      AS month,
            COUNT(*)                             AS admissions,
            ROUND(SUM(billing_amount), 2)        AS revenue,
            ROUND(AVG(billing_amount), 2)        AS avg_billing,
            ROUND(AVG(length_of_stay), 2)        AS avg_los
        FROM healthcare_records
        WHERE date_of_admission IS NOT NULL
        GROUP BY month
        ORDER BY month
    """)

def get_yearly_trend() -> pd.DataFrame:
    return query_df("""
        SELECT
            SUBSTR(date_of_admission, 1, 4)      AS year,
            COUNT(*)                             AS admissions,
            ROUND(SUM(billing_amount), 2)        AS revenue,
            COUNT(DISTINCT medical_condition)    AS conditions_seen
        FROM healthcare_records
        WHERE date_of_admission IS NOT NULL
        GROUP BY year
        ORDER BY year
    """)

def get_condition_summary() -> pd.DataFrame:
    return query_df("""
        SELECT
            medical_condition,
            COUNT(*)                             AS count,
            ROUND(AVG(billing_amount), 2)        AS avg_billing,
            ROUND(AVG(length_of_stay), 2)        AS avg_los,
            SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END) AS abnormal
        FROM healthcare_records
        GROUP BY medical_condition
        ORDER BY count DESC
    """)

def get_recent_records(n: int = 50) -> pd.DataFrame:
    return query_df(f"""
        SELECT
            patient_name, age, gender, medical_condition,
            admission_type, date_of_admission, discharge_date,
            length_of_stay, billing_amount, test_results, hospital
        FROM healthcare_records
        ORDER BY date_of_admission DESC
        LIMIT {n}
    """)

```

File: analytics/patient_analytics.py (Patient Analytics Engine)

```

"""
Patient demographics analytics queries – SQLite-compatible.
"""

import pandas as pd
from database.connection import query_df

def get_age_distribution() -> pd.DataFrame:
    return query_df("""
        SELECT
            age_group,

```

```

        COUNT(*) AS count,
        ROUND(100.0 * COUNT(*) / (SELECT COUNT(*) FROM healthcare_records), 2) AS pct
    FROM healthcare_records
    GROUP BY age_group
    ORDER BY CASE age_group
        WHEN 'Under 18' THEN 1
        WHEN '18-29'   THEN 2
        WHEN '30-44'   THEN 3
        WHEN '45-59'   THEN 4
        WHEN '60-74'   THEN 5
        WHEN '75+'     THEN 6
        ELSE 7
    END
    """)

def get_gender_breakdown() -> pd.DataFrame:
    return query_df("""
        SELECT
            gender,
            COUNT(*) AS count,
            ROUND(AVG(age), 1) AS avg_age,
            ROUND(AVG(billing_amount), 2) AS avg_billing,
            ROUND(100.0 * COUNT(*) / (SELECT COUNT(*) FROM healthcare_records), 2) AS pct
        FROM healthcare_records
        GROUP BY gender
    """)

def get_blood_type_distribution() -> pd.DataFrame:
    return query_df("""
        SELECT blood_type, COUNT(*) AS count
        FROM healthcare_records
        WHERE blood_type IS NOT NULL
        GROUP BY blood_type
        ORDER BY count DESC
    """)

def get_insurance_breakdown() -> pd.DataFrame:
    return query_df("""
        SELECT
            insurance_provider,
            COUNT(*) AS patients,
            ROUND(SUM(billing_amount), 2) AS total_billed,
            ROUND(AVG(billing_amount), 2) AS avg_billed,
            ROUND(100.0 * COUNT(*) / (SELECT COUNT(*) FROM healthcare_records), 2) AS pct
        FROM healthcare_records
        GROUP BY insurance_provider
        ORDER BY patients DESC
    """)

def get_age_gender_heatmap() -> pd.DataFrame:
    return query_df("""
        SELECT
            age_group,
            gender,
            COUNT(*) AS count
        FROM healthcare_records
        GROUP BY age_group, gender
        ORDER BY CASE age_group
            WHEN 'Under 18' THEN 1
            WHEN '18-29'   THEN 2
            WHEN '30-44'   THEN 3
            WHEN '45-59'   THEN 4
            WHEN '60-74'   THEN 5
            WHEN '75+'     THEN 6
            ELSE 7
        END
    """)

def get_age_vs_billing() -> pd.DataFrame:
    return query_df("""

```

```

        SELECT age, billing_amount, medical_condition, gender
        FROM healthcare_records
        LIMIT 3000
    """
)

def get_condition_by_gender() -> pd.DataFrame:
    return query_df("""
        SELECT
            medical_condition,
            gender,
            COUNT(*) AS count
        FROM healthcare_records
        GROUP BY medical_condition, gender
    """)

```

File: analytics/disease_analytics.py (Disease Analytics Engine)

```

"""
Disease analytics - SQLite-compatible queries.
"""

import pandas as pd
from database.connection import query_df

def get_top_conditions(limit: int = 10) -> pd.DataFrame:
    return query_df(f"""
        SELECT
            medical_condition,
            COUNT(*)
            AS total_cases,
            ROUND(AVG(billing_amount), 2)
            AS avg_billing,
            ROUND(AVG(length_of_stay), 2)
            AS avg_los,
            SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END) AS abnormal,
            SUM(CASE WHEN test_results='Normal' THEN 1 ELSE 0 END) AS normal,
            SUM(CASE WHEN test_results='Inconclusive' THEN 1 ELSE 0 END) AS inconclusive,
            ROUND(100.0 * SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END)
            / COUNT(*), 2)
            AS abnormal_rate
        FROM healthcare_records
        GROUP BY medical_condition
        ORDER BY total_cases DESC
        LIMIT {limit}
    """)

def get_condition_trend() -> pd.DataFrame:
    return query_df("""
        SELECT
            SUBSTR(date_of_admission, 1, 7) AS month,
            medical_condition,
            COUNT(*) AS cases
        FROM healthcare_records
        WHERE date_of_admission IS NOT NULL
        GROUP BY month, medical_condition
        ORDER BY month
    """)

def get_condition_by_age_group() -> pd.DataFrame:
    return query_df("""
        SELECT age_group, medical_condition, COUNT(*) AS count
        FROM healthcare_records
        GROUP BY age_group, medical_condition
    """)

def get_medication_distribution() -> pd.DataFrame:
    return query_df("""
        SELECT

```

```

        medication,
        COUNT(*) AS count,
        ROUND(AVG(billing_amount), 2) AS avg_billing,
        medical_condition
    FROM healthcare_records
    GROUP BY medication, medical_condition
    ORDER BY count DESC
    """

def get_medication_summary() -> pd.DataFrame:
    return query_df("""
        SELECT
            medication,
            COUNT(*) AS total_prescribed,
            COUNT(DISTINCT medical_condition) AS conditions_treated,
            ROUND(AVG(billing_amount), 2) AS avg_cost
        FROM healthcare_records
        GROUP BY medication
        ORDER BY total_prescribed DESC
    """)

def get_test_results_by_condition() -> pd.DataFrame:
    return query_df("""
        SELECT medical_condition, test_results, COUNT(*) AS count
        FROM healthcare_records
        GROUP BY medical_condition, test_results
        ORDER BY medical_condition, count DESC
    """)

def get_condition_severity_proxy() -> pd.DataFrame:
    return query_df("""
        SELECT
            medical_condition,
            ROUND(AVG(billing_amount), 2) AS avg_billing,
            ROUND(AVG(length_of_stay), 2) AS avg_los,
            COUNT(*) AS cases,
            ROUND(AVG(age), 1) AS avg_patient_age
        FROM healthcare_records
        GROUP BY medical_condition
        ORDER BY avg_billing DESC
    """)

```

File: analytics/hospital_analytics.py (Hospital Analytics Engine)

```

"""
Hospital operations analytics - SQLite-compatible queries.
"""

import pandas as pd
from database.connection import query_df

def get_hospital_performance() -> pd.DataFrame:
    return query_df("""
        SELECT
            hospital,
            COUNT(*) AS total_admissions,
            ROUND(AVG(billing_amount), 2) AS avg_billing,
            ROUND(SUM(billing_amount), 2) AS total_revenue,
            ROUND(AVG(length_of_stay), 2) AS avg_los,
            COUNT(DISTINCT doctor) AS doctors,
            SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END) AS abnormal_cases,
            ROUND(100.0 * SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END)
                  / COUNT(*), 2) AS abnormal_rate
        FROM healthcare_records
        GROUP BY hospital
    """)

```

```

        ORDER BY total_admissions DESC
    """)

def get_admission_type_breakdown() -> pd.DataFrame:
    return query_df("""
        SELECT
            admission_type,
            COUNT(*) AS count,
            ROUND(AVG(billing_amount), 2) AS avg_billing,
            ROUND(AVG(length_of_stay), 2) AS avg_los,
            ROUND(100.0 * COUNT(*) / (SELECT COUNT(*) FROM healthcare_records), 2) AS pct
        FROM healthcare_records
        GROUP BY admission_type
    """)

def get_los_distribution() -> pd.DataFrame:
    return query_df("""
        SELECT length_of_stay AS los, COUNT(*) AS count
        FROM healthcare_records
        WHERE length_of_stay IS NOT NULL
            AND length_of_stay BETWEEN 1 AND 60
        GROUP BY length_of_stay
        ORDER BY length_of_stay
    """)

def get_los_by_condition() -> pd.DataFrame:
    return query_df("""
        SELECT
            medical_condition,
            ROUND(AVG(length_of_stay), 2) AS avg_los,
            MIN(length_of_stay) AS min_los,
            MAX(length_of_stay) AS max_los
        FROM healthcare_records
        WHERE length_of_stay IS NOT NULL
        GROUP BY medical_condition
        ORDER BY avg_los DESC
    """)

def get_doctor_performance() -> pd.DataFrame:
    return query_df("""
        SELECT
            doctor,
            COUNT(*) AS patients_seen,
            ROUND(AVG(billing_amount), 2) AS avg_billing,
            ROUND(AVG(length_of_stay), 2) AS avg_los,
            COUNT(DISTINCT medical_condition) AS conditions_treated,
            COUNT(DISTINCT hospital) AS hospitals_worked,
            SUM(CASE WHEN test_results='Normal' THEN 1 ELSE 0 END) AS normal_outcomes,
            SUM(CASE WHEN test_results='Abnormal' THEN 1 ELSE 0 END) AS abnormal_outcomes
        FROM healthcare_records
        GROUP BY doctor
        ORDER BY patients_seen DESC
        LIMIT 50
    """)

def get_admission_type_by_condition() -> pd.DataFrame:
    return query_df("""
        SELECT medical_condition, admission_type, COUNT(*) AS count
        FROM healthcare_records
        GROUP BY medical_condition, admission_type
    """)

def get_weekly_admission_pattern() -> pd.DataFrame:
    return query_df("""
        SELECT
            CASE CAST(strftime('%w', date_of_admission) AS INTEGER)
                WHEN 0 THEN 'Sunday'
                WHEN 1 THEN 'Monday'
                WHEN 2 THEN 'Tuesday'
    """)

```

```

        WHEN 3 THEN 'Wednesday'
        WHEN 4 THEN 'Thursday'
        WHEN 5 THEN 'Friday'
        WHEN 6 THEN 'Saturday'
    END AS day_name,
    CAST(strftime('%w', date_of_admission) AS INTEGER) AS day_num,
    COUNT(*) AS admissions
FROM healthcare_records
WHERE date_of_admission IS NOT NULL
GROUP BY day_num, day_name
ORDER BY day_num
""")

def get_room_utilization() -> pd.DataFrame:
    return query_df("""
        SELECT
            room_number,
            COUNT(*) AS usage_count,
            ROUND(AVG(billing_amount), 2) AS avg_billing
        FROM healthcare_records
        WHERE room_number IS NOT NULL AND room_number > 0
        GROUP BY room_number
        ORDER BY usage_count DESC
        LIMIT 20
    """)

```

File: analytics/financial_analytics.py (Financial Analytics Engine)

```

"""
Financial analytics - SQLite-compatible queries.
"""

import pandas as pd
from database.connection import query_df

def get_revenue_summary() -> dict:
    df = query_df("""
        SELECT
            ROUND(SUM(billing_amount), 2) AS total_revenue,
            ROUND(AVG(billing_amount), 2) AS avg_billing,
            ROUND(MIN(billing_amount), 2) AS min_billing,
            ROUND(MAX(billing_amount), 2) AS max_billing
        FROM healthcare_records
    """)
    row = df.iloc[0].to_dict()
    # Compute stddev manually in Python (SQLite has no STDDEV)
    bills_df = query_df("SELECT billing_amount FROM healthcare_records WHERE billing_amount IS NOT NULL")
    import numpy as np
    row["stddev_billing"] = round(float(np.std(bills_df["billing_amount"])), 2)
    return row

def get_revenue_by_condition() -> pd.DataFrame:
    return query_df("""
        SELECT
            medical_condition,
            ROUND(SUM(billing_amount), 2) AS total_revenue,
            ROUND(AVG(billing_amount), 2) AS avg_billing,
            COUNT(*) AS cases
        FROM healthcare_records
        GROUP BY medical_condition
        ORDER BY total_revenue DESC
    """)

def get_revenue_by_insurance() -> pd.DataFrame:

```

```

return query_df("""
    SELECT
        insurance_provider,
        COUNT(*)                AS patients,
        ROUND(SUM(billing_amount), 2) AS total_billed,
        ROUND(AVG(billing_amount), 2) AS avg_billed,
        ROUND(MIN(billing_amount), 2) AS min_billed,
        ROUND(MAX(billing_amount), 2) AS max_billed
    FROM healthcare_records
    GROUP BY insurance_provider
    ORDER BY total_billed DESC
""")

def get_revenue_by_admission_type() -> pd.DataFrame:
    return query_df("""
        SELECT
            admission_type,
            COUNT(*)                AS cases,
            ROUND(SUM(billing_amount), 2) AS total_revenue,
            ROUND(AVG(billing_amount), 2) AS avg_revenue
        FROM healthcare_records
        GROUP BY admission_type
        ORDER BY total_revenue DESC
    """)

def get_monthly_revenue_trend() -> pd.DataFrame:
    return query_df("""
        SELECT
            SUBSTR(date_of_admission, 1, 7) AS month,
            ROUND(SUM(billing_amount), 2) AS revenue,
            COUNT(*) AS admissions,
            ROUND(AVG(billing_amount), 2) AS avg_billing
        FROM healthcare_records
        WHERE date_of_admission IS NOT NULL
        GROUP BY month
        ORDER BY month
    """)

def get_billing_distribution() -> pd.DataFrame:
    return query_df("""
        SELECT billing_amount, medical_condition, insurance_provider
        FROM healthcare_records
        WHERE billing_amount IS NOT NULL
        LIMIT 5000
    """)

def get_revenue_by_hospital() -> pd.DataFrame:
    return query_df("""
        SELECT
            hospital,
            COUNT(*)                AS admissions,
            ROUND(SUM(billing_amount), 2) AS total_revenue,
            ROUND(AVG(billing_amount), 2) AS avg_billing
        FROM healthcare_records
        GROUP BY hospital
        ORDER BY total_revenue DESC
    """)

def get_high_value_cases(threshold: float = 30000) -> pd.DataFrame:
    return query_df(f"""
        SELECT
            patient_name, age, gender, medical_condition,
            hospital, doctor, billing_amount, length_of_stay,
            insurance_provider, admission_type, test_results
        FROM healthcare_records
        WHERE billing_amount >= {threshold}
        ORDER BY billing_amount DESC
        LIMIT 200
    """)

```

File: analytics/ml_insights.py (Machine Learning Analytics Engine)

```
"""
ML-powered insights: clustering, risk scoring, test result prediction.
Uses scikit-learn on data pulled from MySQL.
"""

import warnings
warnings.filterwarnings("ignore")

import pandas as pd
import numpy as np
from database.connection import query_df

# — Raw data for ML —————
def get_ml_data() -> pd.DataFrame:
    return query_df("""
        SELECT
            id, age, gender, blood_type, medical_condition,
            insurance_provider, billing_amount, room_number,
            admission_type, length_of_stay, medication, test_results
        FROM healthcare_records
        WHERE billing_amount IS NOT NULL
            AND length_of_stay IS NOT NULL
            AND length_of_stay > 0
    """)

# — K-Means Clustering —————
def run_clustering(n_clusters: int = 4) -> pd.DataFrame:
    """Cluster patients into risk/cost segments."""
    from sklearn.preprocessing import LabelEncoder, StandardScaler
    from sklearn.cluster import KMeans
    from sklearn.decomposition import PCA

    df = get_ml_data()

    features = ["age", "billing_amount", "length_of_stay"]
    le = LabelEncoder()
    df["gender_enc"] = le.fit_transform(df["gender"].astype(str))
    df["admission_enc"] = le.fit_transform(df["admission_type"].astype(str))
    df["condition_enc"] = le.fit_transform(df["medical_condition"].astype(str))

    feature_cols = features + ["gender_enc", "admission_enc", "condition_enc"]
    X = df[feature_cols].fillna(0)

    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)

    km = KMeans(n_clusters=n_clusters, random_state=42, n_init=10)
    df["cluster"] = km.fit_predict(X_scaled)

    cluster_labels = {
        0: "Low Cost / Short Stay",
        1: "High Cost / Long Stay",
        2: "Emergency / Critical",
        3: "Elective / Routine",
    }
    df["cluster_label"] = df["cluster"].map(
        lambda c: cluster_labels.get(c, f"Segment {c+1}")
    )

    # PCA for 2-D visualization
    pca = PCA(n_components=2, random_state=42)
    coords = pca.fit_transform(X_scaled)
    df["pca_x"] = coords[:, 0]
    df["pca_y"] = coords[:, 1]

    return df[["cluster_label", "pca_x", "pca_y"]]
```



```

        "id", "age", "billing_amount", "length_of_stay", "medical_condition",
        "gender", "admission_type", "cluster", "cluster_label", "pca_x", "pca_y"
    ]]

def get_cluster_summary(cluster_df: pd.DataFrame) -> pd.DataFrame:
    return (
        cluster_df.groupby("cluster_label")
        .agg(
            patients = ("id", "count"),
            avg_age = ("age", "mean"),
            avg_billing = ("billing_amount", "mean"),
            avg_los = ("length_of_stay", "mean"),
        )
        .round(2)
        .reset_index()
        .sort_values("avg_billing", ascending=False)
    )

# — Test Result Predictor —————
def run_test_result_classification():
    """
    Predict test result (Normal / Abnormal / Inconclusive)
    using a Random Forest. Returns model metrics + feature importance.
    """
    from sklearn.ensemble import RandomForestClassifier
    from sklearn.model_selection import train_test_split
    from sklearn.preprocessing import LabelEncoder
    from sklearn.metrics import (classification_report,
                                 confusion_matrix, accuracy_score)

    df = get_ml_data()
    df = df[df["test_results"].isin(["Normal", "Abnormal", "Inconclusive"])]

    le_map = {}
    for col in ["gender", "admission_type", "medical_condition", "medication",
               "insurance_provider", "blood_type"]:
        le = LabelEncoder()
        df[col + "_enc"] = le.fit_transform(df[col].astype(str))
        le_map[col] = le

    feature_cols = [
        "age", "billing_amount", "length_of_stay",
        "gender_enc", "admission_type_enc", "medical_condition_enc",
        "medication_enc", "insurance_provider_enc",
    ]
    X = df[feature_cols].fillna(0)
    y = df["test_results"]

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42, stratify=y
    )

    clf = RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)
    clf.fit(X_train, y_train)
    y_pred = clf.predict(X_test)

    acc = round(accuracy_score(y_test, y_pred) * 100, 2)
    report = classification_report(y_test, y_pred, output_dict=True)
    cm = confusion_matrix(y_test, y_pred, labels=["Normal", "Abnormal", "Inconclusive"])
    fi = pd.DataFrame({
        "feature": feature_cols,
        "importance": clf.feature_importances_
    }).sort_values("importance", ascending=False)

    return {
        "accuracy": acc,
        "report": report,
        "confusion_matrix": cm,
        "labels": ["Normal", "Abnormal", "Inconclusive"],
        "feature_importance": fi,
        "model": clf,
    }

```

```

# — Risk Score —————
def compute_risk_scores() -> pd.DataFrame:
    """
    Composite risk score per patient record based on:
    - Age (older = higher risk)
    - Billing amount (higher = more complex)
    - Length of stay (longer = more serious)
    - Abnormal test result (+risk)
    - Emergency admission (+risk)
    Returns DataFrame with risk_score (0-100) and risk_tier.
    """
    df = get_ml_data()

    # Normalise 0-1
    def norm(s):
        mn, mx = s.min(), s.max()
        return (s - mn) / (mx - mn + 1e-9)

    score = (
        0.30 * norm(df["age"]) +
        0.30 * norm(df["billing_amount"]) +
        0.25 * norm(df["length_of_stay"]) +
        0.10 * (df["test_results"] == "Abnormal").astype(float) +
        0.05 * (df["admission_type"] == "Emergency").astype(float)
    )
    df["risk_score"] = (score * 100).round(1)
    df["risk_tier"] = pd.cut(
        df["risk_score"],
        bins=[0, 33, 66, 100],
        labels=["Low Risk", "Moderate Risk", "High Risk"]
    ).astype(str)

    return df[["id", "age", "billing_amount", "length_of_stay",
               "medical_condition", "gender", "admission_type",
               "test_results", "risk_score", "risk_tier"]]

```
